SURVIVING THE FUTURE

Crisis, Climate, and Human Resilience

A Data-Driven, Academically Grounded Guide to Living Through the Crises Caused by Global Warming

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PREFACE: WHY FUTURE GLOBAL WARMING IS CERTAIN

Global warming is not a hypothesis; it is an observed, measured, and physically explained reality. The certainty of future warming rests on three pillars: (1) the physics of the greenhouse effect, (2) the observed accumulation of greenhouse gases in the atmosphere, and (3) the inertia of the climate system. Even if all emissions stopped today, committed warming from past emissions would continue for decades. This book is written to be debated in academic spaces: every major claim is tied to peer-reviewed literature, data, and expert statements.

The Three Pillars of Certainty

- **Physics:** Greenhouse gases absorb outgoing infrared radiation. This is established physics, tested for over a century (Arrhenius, 1896; modern radiative transfer models).
- **Observation:** Atmospheric CO2 has risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). Global mean temperature has risen ~1.1-1.2 C. These are direct measurements.
- **Inertia:** The ocean stores heat; ice sheets respond slowly. Even with zero emissions, warming continues for decades (committed warming).

Data underpinning the discussion is summarized as follows:

Table P.1 Key global climate indicators (2024)

Indicator	Value	Source
Global mean temperature rise since 1850-1900	~1.1-1.2 C	IPCC (2023)
CO2 concentration (2024)	~420 ppm	NOAA (2024)
Sea level rise since 1900	~20-25 cm	IPCC (2023)
Arctic sea ice decline	~13% per decade	NASA
People affected by climate disasters (annual)	~100 million+	IFRC
Economic damage from climate disasters (annual)	~$200-300 billion	World Bank

The expert consensus underpinning this book is summarized as follows.

Dr. James Hansen, former NASA GISS Director (2013 op-ed/2016 paper): 'Sea level rise of several meters could be expected with 2°C warming — 2°C would have consequences that can be described as disastrous.'

Prof. Katharine Hayhoe, Texas Tech Chief Scientist (2014, post-IPCC AR5): 'Climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.'

Dr. Michael Mann, Penn State climatologist (2004/2021): 'There is about as much scientific consensus about human-caused climate change as there is about gravity — multiple independent lines of evidence converge.'

Dr. Friederike Otto, Imperial College / World Weather Attribution co-founder (2021): 'The Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.'

Prof. Johan Rockström, Potsdam Institute & Stockholm Resilience Centre (2009 Nature; 2023 update): 'We define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.'

Preface (Extended)

This extended preface provides additional context for the reader. The book is organized to move from foundational science to practical action, ensuring that every reader can find relevant material. The structure reflects the complexity of the climate crisis and the need for multi-level responses.

Additional context: The climate crisis demands sustained attention and coordinated action across all sectors of society. This book aims to equip readers with the knowledge and tools needed to contribute meaningfully to that effort.

CHAPTER 1

INTRODUCTION: THE WARMING WORLD

Theme: the warming world and the coming crises

This chapter examines the warming world and the coming crises. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

1.1 Why This Book

As Zeke Hausfather (2023) argues, warming now outpaces any 2,000-year period, and 2023 broke records by a wide margin.

Hausfather's analyses put recent warming in the context of the last two thousand years and track how 2023 broke records by an unusual margin. The record is partly amplified by natural variability on top of the trend. He argues that model projections have held up well against observations. Applied to why this book, the argument sets a test: does the measure still work when capacity is stretched?

The case for this book is narrow and deliberately so. It does not argue that the future is known, only that the physical direction of warming is settled while the consequences remain a matter of decision, capacity and timing (IPCC, 2023). Most public argument treats those as one question, which is why debate oscillates between fatalism and delay.

What follows from the distinction is a different kind of book. Rather than forecasting a single future, it sets out what is already measured, what is plausibly projected, and what can still be protected. Each chapter separates those registers so that a reader can see where the evidence is strong, where it is conditional and where it runs out.

Success is not measured by agreement. It is measured by whether a reader can take a claim, check it against the cited source, and decide what to do in a household, a city or an institution. That is a lower ambition than prediction, and a more useful one.

1.2 What 'Survive' Really Means

Zeke Hausfather (2023) puts it plainly: warming now outpaces any 2,000-year period, and 2023 broke records by a wide margin.

Hausfather's analyses put recent warming in the context of the last two thousand years and track how 2023 broke records by an unusual margin. The record is partly amplified by natural variability on top of the trend. He argues that model projections have held up well against observations. For what 'survive' really means, the bearing is direct: the constraint is institutional as much as physical.

'Survive' is used here in a plain sense: to keep the essentials of a life intact through disruption — water, food, shelter, health, income and the social ties that make recovery possible.

That definition rules out two common shortcuts. One treats survival as individual stockpiling, which fails when systems fail together. The other treats it as purely collective, which leaves households unprepared for the days before help arrives. The disaster record shows that both matter, and that the difference between a bad week and a permanent loss is usually local capacity (UNDRR, 2015).

Survival, on this account, is not a bunker. It is a set of arrangements — some physical, some institutional, some social — that keep options open when conditions change faster than expected.

1.3 How to Use This Book

As Zeke Hausfather (2023) argues, warming now outpaces any 2,000-year period, and 2023 broke records by a wide margin.

Hausfather's analyses put recent warming in the context of the last two thousand years and track how 2023 broke records by an unusual margin. The record is partly amplified by natural variability on top of the trend. He argues that model projections have held up well against observations. Read against how to use this book, the point is that intention without capacity changes little.

The book can be read straight through or used as a reference. Chapters 1-3 establish the physical and institutional baseline; chapters 4-13 work through the main impact domains; chapters 14-21 turn to knowledge, preparedness and governance; chapters 22-23 examine regional cases; chapter 24 collects the practical material.

Each chapter follows the same architecture: an opening claim, four analytical sections, a case study and a scenario table. The tables can be compared across chapters, and references are consolidated in Appendix D so that a claim can be traced without hunting through footnotes.

Readers in a hurry can start with the scenario table at the end of a chapter and work backwards. Readers using the book for teaching can pair the case studies with the checklists in chapter 24. Neither route requires accepting the argument whole.

1.4 The Scope of the Crisis

As Amitav Ghosh (2016) argues, the climate crisis is a crisis of imagination — we have struggled to narrate what we must act upon.

Ghosh argues that the literary imagination has struggled to represent a crisis of this scale. Inability to narrate is part of why action lags. The claim is cultural and complements, rather than replaces, empirical analysis. The implication for the scope of the crisis is a question of sequence: what has to be in place before the event, not after it.

The scope is wide but not unlimited. The book deals with climate-driven risks to human welfare — heat, water, food, health, shelter, displacement, conflict and the fiscal capacity to respond — and with the measures that reduce them.

It does not attempt a full account of every Earth-system change, nor a complete economics of decarbonisation. Those are treated where they bear on survival and cited rather than summarised at length.

The boundary is deliberate: a book that tried to cover everything would explain nothing in usable detail. What is claimed here is narrower — that the risks are documented, the responses are known, and the gap between them is a decision rather than a mystery.

1.5 Case Study: Introduction — The Warming World

One documented episode makes the argument concrete: Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people (WMO, UN OCHA). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. Within public communication about climate risk, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). As far as public communication about climate risk is concerned, the episode works as a readiness test rather than a weather story.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

1.6 Future Scenarios and Predictions: Introduction — The Warming World

For the warming world as a whole, the near term is already measured in centimeters and days: sea level is on track to rise 0.2–0.3 m by 2040 and 0.4–0.6 m by 2060 against 2000, while neighborhoods below 2–3 m above sea level in Jakarta, Semarang, Surabaya, New Orleans, Dhaka and small island capitals face monthly tidal flooding even without storms.

By 2050, an additional 0.8–1.0 °C pushes compound crises — heat, floods and harvest failures arriving together — exposing 20–40% more people than in 2020 (World Bank, 2022); by 2100, committed sea-level rise of 0.6–1.0 m redraws coastlines and threatens fishing villages and ports even if emissions fall after mid-century (Oppenheimer et al., 2019; IPCC, 2023).

The response is layered: raise houses on stilts, restore mangroves as living seawalls, secure freshwater against salt intrusion, and plan staged relocation with jobs rather than sudden displacement.

Table 1.1 Future scenarios and predictions for the warming world and the coming crises (2030-2100)

Horizon	What changes for this chapter
2030	warming ~1.5 °C; sea level +0.10–0.15 m vs 2000
2050	~2.0–2.4 °C; 20–40% more people exposed
2100	0.6–1.0 m sea level; coastlines redrawn

Watch three local signals — annual high-tide flooding days, the depth of the saltwater front in wells, and the retreat of the nearest mangrove belt; global temperature and sea level are high-confidence, but which street floods first depends on drainage and planning (IPCC, 2022).

CHAPTER 2

THE PHYSICAL SCIENCE OF GLOBAL WARMING

Theme: the physical science of global warming and its history

The subject of this chapter is the physical science of global warming and its history. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

2.1 The Greenhouse Effect

Michael Mann (2004) puts it plainly: there is about as much scientific consensus about human-caused climate change as there is about gravity — multiple independent lines of evidence converge.

Mann's statement reflects his work on the 'hockey stick' reconstruction (Mann et al., 1999) and decades defending consensus against misinformation. 'About as much consensus as gravity' is a deliberate analogy to convey that multiple independent lines — surface thermometers, satellites, ocean heat, ice cores, and models — converge on human causation. 'Multiple independent lines of evidence converge' means no single study carries the conclusion; rather, replication across methods and groups reduces uncertainty. Mann points to the role of peer review, open data, and replication in building this confidence. The statement also rebuts false-balance media framing that equates minority dissent with substantive scientific debate. Understanding this convergence helps readers evaluate claims about the greenhouse effect by looking for consistency across datasets, not single outliers. For the physical science of global warming and its history it justifies treating anthropogenic warming as a settled premise for risk planning. For the greenhouse effect, the practical consequence is to name the decision that would have to change.

How a problem is framed decides what counts as a solution, and the greenhouse effect is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). In the work of the physics community and observation networks, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on the greenhouse effect is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: Arctic sea ice declines by ~13% per decade (NASA, 2024). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how the physics community and observation networks should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response

is a stated limitation rather than a confident sentence. For the physics community and observation networks, that is the difference between a plan that can be defended and one that merely exists.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within the physics community and observation networks, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Not all statements about the greenhouse effect carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. As far as the physics community and observation networks is concerned, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The news cycle supplies a concrete marker: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). It matters less as a record than as evidence that the baseline has already moved. In the work of the physics community and observation networks, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of the physics community and observation networks, the operational reading is to record which limit applies to each decision before the decision is taken.

2.2 Greenhouse Gases and Their Sources

As Zeke Hausfather (2023) argues, warming now outpaces any 2,000-year period, and 2023 broke records by a wide margin.

Hausfather's analyses put recent warming in the context of the last two thousand years and track how 2023 broke records by an unusual margin. The record is partly amplified by natural variability on top of the trend. He argues that model projections have held up well against observations. Applied to greenhouse gases and their sources, the argument sets a test: does the measure still work when capacity is stretched?

Evidence on greenhouse gases and their sources rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: Arctic sea ice declines by ~13% per decade (NASA, 2024). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. For the physics community and observation networks, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within the physics community and observation networks, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The strength of the claim about greenhouse gases and their sources varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). A central estimate without its range is not more precise, only less honest. As far as the physics community and observation networks is concerned, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question greenhouse gases and their sources is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: Arctic sea ice declines by ~13% per decade (NASA, 2024). In the work of the physics community and observation networks, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Current reporting adds detail rather than a new direction: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Treated as a single observation it invites over-reading; treated as one of many it is informative. For the physics community and observation networks, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full

range of projections — and those are usually the decisions that are cheap to reverse. For the physics community and observation networks, the question is which of these decisions can be taken before the next event rather than during it.

2.3 Evidence of Warming

As Andrew Dessler (2021) argues, greenhouse physics has been settled since the 1800s; debate is about impacts and choices, not reality.

Dessler points out that the physics of the greenhouse effect has been settled since the nineteenth century. What remains open is the magnitude of impacts and the choices that determine them. Public debate often confuses the two. For evidence of warming, the bearing is direct: the constraint is institutional as much as physical.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within the physics community and observation networks, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Uncertainty in evidence of warming is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). As far as the physics community and observation networks is concerned, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about evidence of warming are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: Arctic sea ice declines by ~13% per decade (NASA, 2024). In the work of the physics community and observation networks, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind evidence of warming is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates

and lowest on localities: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For the physics community and observation networks, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

A recent marker points the same way: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). It is one line in a longer ledger, and the ledger is what matters. Within the physics community and observation networks, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. Within the physics community and observation networks, the operational reading is to record which limit applies to each decision before the decision is taken.

2.4 Temperature Records and Trends

Naomi Oreskes (2010) puts it plainly: doubt was manufactured to delay, and history helps us see the pattern.

Oreskes and Conway documented how a small group of scientists and consultants manufactured doubt on tobacco and then on climate. The strategy targets uncertainty rather than evidence. Knowing the playbook changes how claims are read. Read against temperature records and trends, the point is that intention without capacity changes little.

An honest account of temperature records and trends distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). Numbers carry confidence intervals, and those intervals should travel with them into any decision. As far as the physics community and observation networks is concerned, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Discussion of temperature records and trends usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: Arctic sea ice declines by ~13% per decade (NASA, 2024). Read that way, the question for the physics community and observation networks is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. In the work of the physics community and observation networks, the

implication is a shorter list of decisions than the analysis suggests.

The evidence base for temperature records and trends is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. For the physics community and observation networks, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). Within the physics community and observation networks, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The news cycle supplies a concrete marker: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Set beside the record, the episode indicates direction without fixing the rate. As far as the physics community and observation networks is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. As far as the physics community and observation networks is concerned, the question is which of these decisions can be taken before the next event rather than during it.

2.5 Case Study: The Physical Science of Global Warming

A single episode illustrates the mechanism: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. As far as the physics community and observation networks is concerned, the record supplies both the warning and the timetable.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which the physics community and observation networks controls.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

2.6 Future Scenarios and Predictions: The Physical Science of Global Warming

For the physical science of warming, the next two decades are written in measurable thresholds: global mean temperature reaches about 1.5 °C by 2030 and CO2 about 430 ppm, and by 2040 the Arctic sees 10–20 ice-free summer days a year instead of roughly 5 today, while ocean heat content climbs about 15% and feeds stronger storms.

By 2050 the energy already stored in the ocean commits the planet to further warming even if emissions fall, and by 2100 a high-emissions path yields 3.6–4.4 °C with Greenland melt adding 0.3–0.5 m to sea level (IPCC, 2023); the physics is among the most confident parts of climate science.

Adaptation here means better measurement: expand satellite and ocean-buoy coverage, keep citizen temperature and rainfall logs, and translate the numbers into local heat and sea-level dashboards.

Table 2.1 Future scenarios and predictions for the physical science of global warming and its history (2030-2100)

Horizon	What changes for this chapter
2030	1,5 °C; CO2 ~430 ppm
2050	ocean heat +~15%; committed warming
2100	3,6–4,4 °C high path; Greenland +0,3–0,5 m

Watch the annual CO2 reading at Mauna Loa, ocean heat content in the nearest basin, and the date of the Arctic sea-ice minimum; these are high-confidence global indicators, while regional storm counts remain less certain (IPCC, 2021).

CHAPTER 3

CLIMATE SYSTEMS, FEEDBACKS AND MONITORING

Theme: climate systems, feedback mechanisms and monitoring

This chapter deals with climate systems, feedback mechanisms and monitoring. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

3.1 The Climate System

As Johan Rockström (2009) argues, we define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.

Rockström's statement originates from the 2009 Nature paper 'A safe operating space for humanity' and its 2023 update that now finds six of nine boundaries transgressed. 'Planetary boundaries' are quantitative thresholds for climate, biosphere, freshwater, nutrients, and other Earth-system processes that define a Holocene-like stability zone. 'Operate safely' means staying within these thresholds reduces risk of abrupt, irreversible shifts. 'Six of nine transgressed' indicates humanity has left the safe zone on climate, biodiversity, land, freshwater, biogeochemical flows, and novel entities, far outside preindustrial variability. The update emphasizes interactions — transgressing one boundary tightens others, such as deforestation amplifying carbon release. The concept reframes the climate system from isolated sectoral problem to systemic interdependence. For climate systems, it implies solutions must be integrated across energy, food, and ecosystem management rather than siloed. In the terms of the climate system, the claim shifts attention from prediction to preparation.

Evidence on the climate system rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: Arctic sea ice declines by ~13% per decade (NASA, 2024). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. As far as monitoring agencies and forecasting services is concerned, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For that reason, average-based targets can be met while the most exposed end up worse off. In the work of monitoring agencies and forecasting services, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For monitoring agencies and forecasting services, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: Arctic sea ice declines by ~13% per decade (NASA, 2024). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within monitoring agencies and forecasting services, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The news cycle supplies a concrete marker: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). The item is useful because it fixes a date to a trend that otherwise drifts. For monitoring agencies and forecasting services, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as monitoring agencies and forecasting services is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

3.2 Atmosphere and Ocean Circulation

As Carl Folke (2010) argues, resilience thinking reframes crisis as reorganization, focusing on thresholds, diversity, and learning.

Folke's resilience thinking treats crises as reorganisation rather than collapse. Thresholds, diversity and learning determine whether a system absorbs a shock or flips into a less desirable state. Management therefore aims at keeping options open. For atmosphere and ocean circulation, the practical consequence is to name the decision that would have to change.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. In the work of monitoring agencies and forecasting services, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to

perform better for everyone else as well.

Institutional performance in atmosphere and ocean circulation depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For monitoring agencies and forecasting services, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: Arctic sea ice declines by ~13% per decade (NASA, 2024). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within monitoring agencies and forecasting services, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

The evidence base behind atmosphere and ocean circulation is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). As far as monitoring agencies and forecasting services is concerned, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Current reporting adds detail rather than a new direction: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). What the report adds is timing, not a new mechanism. Within monitoring agencies and forecasting services, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. In the work of monitoring agencies and forecasting services, the question is which of these decisions can be taken before the next event rather than during it.

3.3 Positive and Negative Feedback Loops

Pierre Friedlingstein (2020) puts it plainly: land and ocean sinks are weakening under warming, so we cannot count on nature to clean up after us.

Friedlingstein and colleagues measure how much carbon land and ocean sinks absorb each year. Both sinks weaken under warming, so a larger share of emissions stays in the atmosphere. The measurement carries uncertainty but the direction is consistent. Applied to positive and negative feedback loops, the argument sets a test: does the measure still work when capacity is stretched?

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For monitoring agencies and forecasting services, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: Arctic sea ice declines by ~13% per decade (NASA, 2024). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within monitoring agencies and forecasting services, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The evidence base for positive and negative feedback loops is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. As far as monitoring agencies and forecasting services is concerned, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Averages describe the size of the problem, not who carries it. Decisions taken by monitoring agencies and forecasting services — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. In the work of monitoring agencies and forecasting services, the test is whether the arrangement survives contact with a bad week.

A recent marker points the same way: Antarctic sea ice hit record winter low in Sept 2023, 1 million km2 below average (NSIDC, NASA). The event is best read as a measurement, not as a verdict. As far as monitoring agencies and forecasting services is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. For monitoring agencies and forecasting services, the operational reading is to record which limit applies to each decision before the decision is taken.

3.4 Tipping Points and Abrupt Change

As Johan Rockström (2009) argues, we define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.

Rockström's statement originates from the 2009 Nature paper 'A safe operating space for humanity' and its 2023 update that now finds six of nine boundaries transgressed. 'Planetary boundaries' are quantitative thresholds for climate, biosphere, freshwater, nutrients, and other Earth-system processes that define a Holocene-like stability zone. 'Operate safely' means staying within these thresholds reduces risk of abrupt, irreversible shifts. 'Six of nine transgressed' indicates humanity has left the safe zone on climate, biodiversity, land, freshwater, biogeochemical flows, and novel entities, far outside preindustrial variability. The update emphasizes interactions — transgressing one boundary tightens others, such as deforestation amplifying carbon release. The concept reframes tipping points and abrupt change from isolated sectoral problem to systemic interdependence. For climate systems, it implies solutions must be integrated across energy, food, and ecosystem management rather than siloed. For tipping points and abrupt change, the bearing is direct: the constraint is institutional as much as physical.

The economics of tipping points and abrupt change are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: Arctic sea ice declines by ~13% per decade (NASA, 2024). Within monitoring agencies and forecasting services, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

The record on tipping points and abrupt change is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how monitoring agencies and forecasting services should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. As far as

monitoring agencies and forecasting services is concerned, that is the difference between a plan that can be defended and one that merely exists.

The aggregate picture hides the distribution, and in tipping points and abrupt change the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). In the work of monitoring agencies and forecasting services, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: Arctic sea ice declines by ~13% per decade (NASA, 2024). What the record does show is that authority, money and information have to move together; any one without the others stalls. For monitoring agencies and forecasting services, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. In the work of monitoring agencies and forecasting services, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. Within monitoring agencies and forecasting services, the question is which of these decisions can be taken before the next event rather than during it.

3.5 Case Study: Climate Systems, Feedbacks and Monitoring

The case that anchors this chapter is documented and recent: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. In the work of monitoring agencies and forecasting services, the sequence is worth reconstructing before the next season rather than after it.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted

savings — a sequence monitoring agencies and forecasting services can measure in its own records.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

3.6 Future Scenarios and Predictions: Climate Systems, Feedbacks and Monitoring

For climate systems and feedbacks, the near term is about thresholds that can switch: by 2040 permafrost thaw releases 10–20% more methane, Amazon dieback risk rises where deforestation passes 20–25%, and the Atlantic overturning circulation slows 15–25%, while early-warning sensors in 30% of vulnerable basins could still catch tipping signals.

By 2050 several feedbacks begin to reinforce each other — fire-driven forest loss, soil carbon release and weaker carbon sinks — and by 2100 a high-emissions path risks passing tipping points that are effectively irreversible on human timescales (IPCC, 2022; Steffen et al., 2018).

The response is monitoring plus precaution: deploy permafrost and forest sensors, link community rain gauges into national networks, and set trigger points that automatically escalate policy when a threshold is approached.

Table 3.1 Future scenarios and predictions for climate systems, feedback mechanisms and monitoring (2030-2100)

Horizon	What changes for this chapter
2030	methane +10–20%; AMOC −15–25%
2050	feedbacks reinforce; sinks weaken
2100	tipping risk rises sharply (high emissions)

Watch permafrost borehole temperatures, the Amazon's dry-season greenness, and the strength of the Atlantic overturning; these indicators carry medium confidence, and their interactions are the least certain part of the picture (IPCC, 2021).

CHAPTER 4

EXTREME HEAT AND URBAN HEALTH

Theme: extreme heat, heat waves and urban heat islands

What follows concerns extreme heat, heat waves and urban heat islands. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

4.1 Heat Wave Definition and Trends

As Friederike Otto (2021) argues, the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that heat wave definition and trends extremes are not random bad luck but part of a detectable trend. For extreme heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. The implication for heat wave definition and trends is a question of sequence: what has to be in place before the event, not after it.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within city health services and labour policy, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Anyone deciding about heat wave definition and trends is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as city health services and labour policy is concerned, the test of a plan is not whether it is optimal for one forecast but whether it survives

several. Decisions taken now will be judged by events that were not in the forecast.

Most disagreements about heat wave definition and trends are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of city health services and labour policy, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

Uncertainty in heat wave definition and trends is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For city health services and labour policy, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The news cycle supplies a concrete marker: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Treated as a single observation it invites over-reading; treated as one of many it is informative. Within city health services and labour policy, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. Within city health services and labour policy, the operational reading is to record which limit applies to each decision before the decision is taken.

4.2 Physiology of Heat Stress

Friederike Otto (2021) puts it plainly: the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that physiology of heat stress extremes are not random bad luck but part of a detectable

trend. For extreme heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. In the terms of physiology of heat stress, the claim shifts attention from prediction to preparation.

The relevant question for physiology of heat stress is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as city health services and labour policy is concerned, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

Discussion of physiology of heat stress usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Read that way, the question for city health services and labour policy is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. In the work of city health services and labour policy, the implication is a shorter list of decisions than the analysis suggests.

An honest account of physiology of heat stress distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Numbers carry confidence intervals, and those intervals should travel with them into any decision. For city health services and labour policy, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~7 million air-pollution deaths occur annually (WHO, 2023). Within city health services and labour policy, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

Current reporting adds detail rather than a new direction: India's wheat yields fell 10% in 2022 heatwave, prompting export ban affecting global prices (FAO, India IMD). It is one line in a longer ledger, and the ledger is what matters. As far as city health services and labour policy is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as city health services and labour policy is concerned, the question is which of these decisions can be taken before the next event rather than during it.

4.3 Heat Illness Spectrum

As Friederike Otto (2021) argues, the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that heat illness spectrum extremes are not random bad luck but part of a detectable trend. For extreme heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. For heat illness spectrum, the practical consequence is to name the decision that would have to change.

How a problem is framed decides what counts as a solution, and heat illness spectrum is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of city health services and labour policy, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

Not all statements about heat illness spectrum carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. For city health services and labour policy, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale

(UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~7 million air-pollution deaths occur annually (WHO, 2023). Within city health services and labour policy, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Any assessment of heat illness spectrum will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. As far as city health services and labour policy is concerned, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

A recent marker points the same way: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). Set beside the record, the episode indicates direction without fixing the rate. In the work of city health services and labour policy, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of city health services and labour policy, the operational reading is to record which limit applies to each decision before the decision is taken.

4.4 Vulnerable Populations

As Friederike Otto (2021) argues, the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

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The strength of the claim about vulnerable populations varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). A central estimate without its range is not more precise, only less honest. For city health services and labour policy, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~7 million air-pollution deaths occur annually (WHO, 2023). Within city health services and labour policy, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). None of those signals is a perfect measure, but together they indicate direction. As far as city health services and labour policy is concerned, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

It helps to ask what question vulnerable populations is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). In the work of city health services and labour policy, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

The news cycle supplies a concrete marker: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Read against the longer record, it is an illustration rather than a proof. For city health services and labour policy, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. For city health services and labour policy, the question is which of these decisions can be

taken before the next event rather than during it.

4.5 Case Study: Extreme Heat and Urban Health

One documented episode makes the argument concrete: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. For city health services and labour policy, the episode fixes the date at which the argument stops being hypothetical.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence city health services and labour policy can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

4.6 Future Scenarios and Predictions: Extreme Heat and Urban Health

For extreme heat and urban health, the near term is counted in dangerous days: by 2040 days above 35 °C double to 60–90 a year in Delhi, Phoenix, Jakarta and Lagos, nights above 28 °C triple, and heat-stroke emergency visits rise 30–50%, while dense kampung neighborhoods run 5–7 °C hotter than nearby parks.

By 2050, with 0.8–1.0 °C more warming, heat becomes the leading climate killer in many cities and cuts outdoor work hours by 10–15%; by 2100 a high-emissions path adds 20–30 dangerous days a year and makes night-time cooling rare in tropical megacities (IPCC, 2022; Watts et al., 2021).

Cool roofs, street trees every 10 m, night cooling shelters, shifted work hours (06–10 and 16–19) and hydration points with heat-alert SMS are the cheapest first line of defense.

Table 4.1 Future scenarios and predictions for extreme heat, heat waves and urban heat islands (2030-2100)

Horizon	What changes for this chapter
2030	35 °C days +30–50%; ER visits +30%
2050	60–90 hot days; work hours −10–15%
2100	+20–30 dangerous days; nights rarely cool

Watch the count of nights above 28 °C, ambulance calls during heat alerts, and the temperature gap between your block and the nearest park; the heat trend is high-confidence, but neighborhood-level exposure depends on housing and shade.

CHAPTER 5

WATER SCARCITY AND DROUGHT

Theme: water scarcity and drought

This chapter examines water scarcity and drought. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

5.1 Global Water Cycle

Peter Gleick (2020) puts it plainly: water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. Read against global water cycle, the point is that intention without capacity changes little.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). For that reason, average-based targets can be met while the most exposed end up worse off. For water utilities and irrigated agriculture, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. Within water utilities and irrigated agriculture, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~2 billion people lack safe water (WHO & UNICEF, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. As far as water utilities and irrigated agriculture is concerned, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). None of those signals is a perfect measure, but together they indicate direction. In the work of water utilities and irrigated agriculture, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: California's atmospheric rivers in winter 2022-23 ended drought but caused $4.6B flood damages (NOAA, CA DWR). What the report adds is timing, not a new mechanism. As far as water utilities and irrigated agriculture is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For water utilities and irrigated agriculture, the operational reading is to record which limit applies to each decision before the decision is taken.

5.2 Drought Types and Indices

As Raj Patel (2022) argues, hunger reflects power, not scarcity — climate shocks deepen those power gaps.

Patel argues that hunger reflects power and access rather than global scarcity. Climate shocks deepen existing gaps in entitlement. The framing shifts attention from production to distribution. The implication for drought types and indices is a question of sequence: what has to be in place before the event, not after it.

Institutional performance in drought types and indices depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~800 million people face food insecurity (FAO, 2023). Within water utilities and irrigated agriculture, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~2 billion people lack safe water (WHO & UNICEF, 2023). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. As far as water utilities and irrigated agriculture is concerned, the relevant question is not whether prevention pays but who captures the

return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Anyone deciding about drought types and indices is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). In the work of water utilities and irrigated agriculture, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~800 million people face food insecurity (FAO, 2023). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. For water utilities and irrigated agriculture, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Current reporting adds detail rather than a new direction: Rhine River fell to critical lows in Aug 2022, disrupting 80% of German inland shipping (German Federal Waterways, Reuters). The event is best read as a measurement, not as a verdict. In the work of water utilities and irrigated agriculture, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within water utilities and irrigated agriculture, the question is which of these decisions can be taken before the next event rather than during it.

5.3 Causes of Water Scarcity

As Peter Gleick (2020) argues, water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. In the terms of causes of water scarcity, the claim shifts attention from prediction to preparation.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~2 billion people lack safe water (WHO & UNICEF, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. As far as water utilities and irrigated agriculture is concerned, the implication

is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for causes of water scarcity is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). In the work of water utilities and irrigated agriculture, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~800 million people face food insecurity (FAO, 2023). Averages describe the size of the problem, not who carries it. Decisions taken by water utilities and irrigated agriculture — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. For water utilities and irrigated agriculture, the practical question is what would have to change first.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~2 billion people lack safe water (WHO & UNICEF, 2023). Within water utilities and irrigated agriculture, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

A recent marker points the same way: California's atmospheric rivers in winter 2022-23 ended drought but caused $4.6B flood damages (NOAA, CA DWR). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. For water utilities and irrigated agriculture, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as water utilities and irrigated agriculture is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

5.4 Agricultural Water Use

Raj Patel (2022) puts it plainly: hunger reflects power, not scarcity — climate shocks deepen those power gaps.

Patel argues that hunger reflects power and access rather than global scarcity. Climate shocks deepen existing gaps in entitlement. The framing shifts attention from production to distribution. For agricultural water use, the practical consequence is to name the decision that would have to change.

Any assessment of agricultural water use will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of water utilities and irrigated agriculture, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in agricultural water use the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~800 million people face food insecurity (FAO, 2023). For water utilities and irrigated agriculture, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~2 billion people lack safe water (WHO & UNICEF, 2023). What the record does show is that authority, money and information have to move together; any one without the others stalls. Within water utilities and irrigated agriculture, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The economics of agricultural water use are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as water utilities and irrigated agriculture is concerned, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

The news cycle supplies a concrete marker: Rhine River fell to critical lows in Aug 2022, disrupting 80% of German inland shipping (German Federal Waterways, Reuters). On its own it proves little; placed against the trend it shows the direction of

travel. Within water utilities and irrigated agriculture, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. In the work of water utilities and irrigated agriculture, the question is which of these decisions can be taken before the next event rather than during it.

5.5 Case Study: Water Scarcity and Drought

A single episode illustrates the mechanism: California's atmospheric rivers in winter 2022-23 ended drought but caused $4.6B flood damages (NOAA, CA DWR). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. Within water utilities and irrigated agriculture, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which water utilities and irrigated agriculture controls.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

5.6 Future Scenarios and Predictions: Water Scarcity and Drought

For water scarcity and drought, the near term shows up in wells and reservoirs: by 2040 renewable water per person falls 25–35% in Java, northern India and the Sahel, groundwater drops 0.5–1.5 m a year, and reservoirs deliver 30% less in dry seasons.

By 2050, drier soils and shorter rainy seasons push 1.5–2.5 billion people into water stress, and by 2100 a high-emissions path cuts river flows 30–50% and dries 40% of shallow wells (IPCC, 2022; WHO & UNICEF, 2023).

Rainwater harvesting on every roof, drip irrigation that cuts use 30–50%, managed aquifer recharge, and village water banks with clear rationing rules buy time.

Table 5.1 Future scenarios and predictions for water scarcity and drought (2030-2100)

Horizon	What changes for this chapter
2030	renewable water −15–25%
2050	1,5–2,5 billion in water stress
2100	river flows −30–50%; 40% wells dry

Watch dry-season reservoir levels, the depth to groundwater in your nearest monitoring well, and the number of tanker deliveries; rainfall trends are medium-confidence, but groundwater decline is directly measurable.

CHAPTER 6

FLOODS AND SEA LEVEL RISE

Theme: floods and sea level rise

The subject of this chapter is floods and sea level rise. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

6.1 Flood Types and Causes

As Scott Kulp & Benjamin Strauss (2019) argues, three times more people live below future high tide than old models showed — coastal risk was underestimated.

Kulp and Strauss corrected coastal elevation data using radar and found that far more land and people sit below future high-tide lines than earlier models showed. Their estimates made exposure visible at the scale of individual settlements. The projections assume stated emissions pathways. For flood types and causes, the bearing is direct: the constraint is institutional as much as physical.

Institutional performance in flood types and causes depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of coastal planning and flood defence, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Most disagreements about flood types and causes are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For coastal planning and flood defence, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind flood types and causes is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Within coastal planning and flood defence, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as coastal planning and flood defence is concerned, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

The news cycle supplies a concrete marker: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). It is one line in a longer ledger, and the ledger is what matters. In the work of coastal planning and flood defence, what matters is not the record but the lead time it implies.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of coastal planning and flood defence, the operational reading is to record which limit applies to each decision before the decision is taken.

6.2 Sea Level Rise Mechanisms

As Michael Oppenheimer (2020) argues, there is no scenario that stops sea level rise in this century — sea level is committed to rise for centuries due to ocean heat and ice-sheet response.

Oppenheimer's statement draws on his role as Lead Author of IPCC SROCC Chapter 4 (2019) on sea-level rise and coastal implications. 'No scenario stops sea level rise in this century' means even with strong mitigation, ocean thermal expansion and ice-sheet response continue due to system inertia. 'Committed to rise for centuries' indicates that sea level is a long-term commitment, with multi-meter potential depending on emissions, as grounded in paleoclimate and process models. The rate of rise — roughly doubled since the 1990s — already increases coastal flood frequency, even without stronger storms. Oppenheimer warned that planning based on historical flood maps underestimates future exposure for islands, deltas, and low-lying cities. For sea level rise mechanisms within floods and sea level rise, the analysis must shift from preventing rise to managing inevitable rise through elevation, protection, accommodation, and managed retreat, with timelines spanning decades to centuries. Read against sea level rise mechanisms, the point is that intention without capacity changes little.

Discussion of sea level rise mechanisms usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022).

Read that way, the question for coastal planning and flood defence is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For coastal planning and flood defence, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for sea level rise mechanisms is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within coastal planning and flood defence, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as coastal planning and flood defence is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of coastal planning and flood defence, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

Current reporting adds detail rather than a new direction: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). Set beside the record, the episode indicates direction without fixing the rate. For coastal planning and flood defence, the item is useful mainly as a date stamp on a trend.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For coastal planning and flood defence, the question is which of these decisions can be taken before the next event rather than during it.

6.3 Coastal Flood Risk

W. Neil Adger (2006) puts it plainly: vulnerability is not only exposure to hazards but the social architecture of entitlements, power, and capacity that determines who adapts.

Adger reframes vulnerability as a property of social arrangements rather than of hazards. Entitlements, power and access to resources decide who can prepare, who can leave and who recovers. The point is structural, not semantic. The implication for coastal flood risk is a question of sequence: what has to be in place before the event, not after it.

The record on coastal flood risk is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how coastal planning and flood defence should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within coastal planning and flood defence, the test is whether the arrangement survives contact with a bad week.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as coastal planning and flood defence is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. In the work of coastal planning and flood defence, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

How a problem is framed decides what counts as a solution, and coastal flood risk is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For coastal planning and flood defence, the practical consequence is that the frame has to be chosen before the plan,

not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). Read against the longer record, it is an illustration rather than a proof. Within coastal planning and flood defence, the report is a reminder that the response is being designed against a target that keeps moving.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. Within coastal planning and flood defence, the operational reading is to record which limit applies to each decision before the decision is taken.

6.4 Riverine and Flash Floods

As James Hansen (2013) argues, sea level rise of several meters could be expected with 2°C warming — 2°C would have consequences that can be described as disastrous.

Hansen's statement originates from his 2016 paper 'Ice melt, sea level rise and superstorms' and a 2013 op-ed that warned about the long-term consequences of 2°C warming. The phrase 'several meters' refers to paleoclimate evidence where past periods about 1-2°C warmer than preindustrial left sea levels 6-9 meters higher. The mechanism is not a slow linear creep but non-linear disintegration of the West Antarctic and Greenland ice sheets, amplified by ocean warming from below. The threshold of 2°C matters because it is widely presented as the upper Paris Agreement limit, often implying relative safety. Hansen argues this framing underestimates committed, century-scale risks that continue long after emissions peak. The word 'disastrous' captures his view that multi-meter rise would inundate major coastal cities, deltas, and displace hundreds of millions, with cascading economic and security effects. For riverine and flash floods within floods and sea level rise, the implication is that near-term temperature targets must be judged against long-term, irreversible sea-level commitments, not just mid-century warming levels. In the terms of riverine and flash floods, the claim shifts attention from prediction to preparation.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as coastal planning and flood defence is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt

(UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. In the work of coastal planning and flood defence, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

It helps to ask what question riverine and flash floods is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For coastal planning and flood defence, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on riverine and flash floods rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~100 million people are affected by climate disasters annually (IFRC, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within coastal planning and flood defence, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). The value of such reports is that they date the change rather than explain it. As far as coastal planning and flood defence is concerned, what matters is not the record but the lead time it implies.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as coastal planning and flood defence is concerned, the question is which of these decisions can be taken before the next event rather than during it.

6.5 Case Study: Floods and Sea Level Rise

The case that anchors this chapter is documented and recent: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. As far as coastal planning and flood defence is concerned, the record supplies both the warning and the timetable.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted

savings — a sequence coastal planning and flood defence can measure in its own records.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

6.6 Future Scenarios and Predictions: Floods and Sea Level Rise

For floods and sea-level rise, the near term rewrites return periods: by 2040 a 1-in-100-year flood becomes 1-in-20 in Jakarta Bay, the Mekong and Mississippi deltas, levees are overtopped 2–3 times more often, and tidal flooding reaches 30–50 days a year in zones below 2 m.

By 2050, 0.2–0.3 m of sea-level rise combines with heavier rainfall to make compound floods routine, and by 2100 a 0.6–1.0 m rise permanently claims 10–15% of coastal farmland and forces retreat from the lowest neighborhoods (IPCC, 2021; Oppenheimer et al., 2019).

Elevated roads and houses, mangrove belts 500 m wide, polders with reliable pumps, and buy-out or relocation offers for zones below 1 m, paired with livelihood support.

Table 6.1 Future scenarios and predictions for floods and sea level rise (2030-2100)

Horizon	What changes for this chapter
2030	1-in-100 → 1-in-30 flood; +0.10 m sea
2050	1-in-100 → 1-in-15; 20–30 tidal days
2100	0.6–1.0 m; 10–15% coastal farmland lost

Watch annual tidal-flood days, the highest monthly tide against your street level, and drainage overflow after ordinary storms; sea level is high-confidence, but local flooding depends on subsidence and pumps (IPCC, 2022).

CHAPTER 7

FOOD SECURITY AND AGRICULTURE

Theme: food security and agriculture under climate change

This chapter deals with food security and agriculture under climate change. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

7.1 Food Systems Overview

As David Lobell (2011) argues, warming since 1980 has already cut wheat and maize yields, masking gains from technology.

Lobell's analysis separated climate effects from technology and management in global crop records. Warming since 1980 reduced wheat and maize yields relative to a counterfactual without it. Adaptation and carbon fertilisation offset part, but not all, of the loss. Applied to food systems overview, the argument sets a test: does the measure still work when capacity is stretched?

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. As far as food systems and rural livelihoods is concerned, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

An honest account of food systems overview distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~100 million people are affected by climate disasters annually (IFRC, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of food systems and rural livelihoods, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~800 million people face food insecurity (FAO, 2023). Averages describe the size of the problem, not who carries it. Decisions taken by food systems and rural livelihoods — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. For food systems and rural livelihoods, the practical question is what would have to change first.

The relevant question for food systems overview is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Within food systems and rural livelihoods, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The news cycle supplies a concrete marker: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). The event is best read as a measurement, not as a verdict. For food systems and rural livelihoods, the item is useful mainly as a date stamp on a trend.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as food systems and rural livelihoods is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

7.2 Crop Yield Impacts

Cynthia Rosenzweig (2020) puts it plainly: the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For crop yield impacts within food security and agriculture under climate change, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. For crop yield impacts, the bearing is direct: the constraint is institutional as much as physical.

Not all statements about crop yield impacts carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~100 million people are affected by climate disasters annually (IFRC, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of food systems and rural livelihoods, the safeguard is to record the source, period and

assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The aggregate picture hides the distribution, and in crop yield impacts the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~800 million people face food insecurity (FAO, 2023). For food systems and rural livelihoods, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Any assessment of crop yield impacts will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. Within food systems and rural livelihoods, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The economics of crop yield impacts are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as food systems and rural livelihoods is concerned, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Current reporting adds detail rather than a new direction: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. Within food systems and rural livelihoods, the report is a reminder that the response is being designed against a target that keeps moving.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of food systems and rural livelihoods, the question is which of these decisions can be taken before the next event rather than during it.

7.3 Heat and Water Stress on Crops

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For heat and water stress on crops within food security and agriculture under climate change, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. Read against heat and water stress on crops, the point is that intention without capacity changes little.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~800 million people face food insecurity (FAO, 2023). For that reason, average-based targets can be met while the most exposed end up worse off. For food systems and rural livelihoods, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). None of those signals is a perfect measure, but together they indicate direction. Within food systems and rural livelihoods, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~100 million people are affected by climate disasters annually (IFRC, 2022). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. As far as food systems and rural livelihoods is concerned, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The strength of the claim about heat and water stress on crops varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~800 million people face food insecurity (FAO, 2023). A central estimate without its range is not more precise, only less honest. In the work of food systems and rural livelihoods, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

A recent marker points the same way: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). On its own it proves little; placed against the trend it shows the direction of travel. As far as food systems and rural livelihoods is concerned, what matters is not the record but the lead time it implies.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For food systems and rural livelihoods, the operational reading is to record which limit applies to each decision before the decision is taken.

7.4 Pests and Diseases

As Tim Wheeler (2013) argues, food shocks travel through prices and nutrition, striking import-dependent poor first.

Wheeler and von Braun trace how food shocks travel through prices, trade and nutrition. Import-dependent poor households absorb the first and deepest losses. Short-term food aid does not prevent the next shock. The implication for pests and diseases is a question of sequence: what has to be in place before the event, not after it.

Anyone deciding about pests and diseases is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Within food systems and rural livelihoods, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~100 million people are affected by climate disasters annually (IFRC, 2022). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. As far as food systems and rural livelihoods is concerned, the relevant question is not whether prevention pays but

who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Uncertainty in pests and diseases is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~800 million people face food insecurity (FAO, 2023). In the work of food systems and rural livelihoods, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. For food systems and rural livelihoods, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

The news cycle supplies a concrete marker: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). Taken alone the episode is weather; taken with the record it is a data point on a slope. In the work of food systems and rural livelihoods, the item is useful mainly as a date stamp on a trend.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within food systems and rural livelihoods, the question is which of these decisions can be taken before the next event rather than during it.

7.5 Case Study: Food Security and Agriculture

One documented episode makes the argument concrete: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. In the work of food systems and rural livelihoods, the sequence is worth reconstructing before the next season rather than after it.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). For food systems and rural livelihoods, the episode works as a readiness test rather than a weather story.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

7.6 Future Scenarios and Predictions: Food Security and Agriculture

For food security and agriculture, the near term appears in yields and prices: by 2040 wheat yields fall 8–15% and maize 10–18% in South Asia and the Sahel, rice drops 5–10% from night heat, and tropical fish catch falls 10–20%, pushing 40–60 million more people into hunger.

By 2050, simultaneous breadbasket failures become plausible and food prices swing 20–35% in bad years; by 2100 a high-emissions path cuts staple output 20–30% without adaptation, while irrigation demand rises 30–40% (Zhao et al., 2017; FAO, 2023).

Heat-tolerant seeds, sowing 2–3 weeks earlier, drip and mulch saving 30% of water, fish ponds and village grain banks smooth the lean months.

Table 7.1 Future scenarios and predictions for food security and agriculture under climate change (2030-2100)

Horizon	What changes for this chapter
2030	wheat −5%; maize −8% in hot zones
2050	price swings 20–35%; 40–60 m hungry
2100	staples −20–30% without adaptation

Watch local yields per hectare, the date of the first reliable rain, and staple prices in the nearest market; crop response to heat is high-confidence, but farm-level losses depend on soil, seed choice and irrigation.

CHAPTER 8

ECOSYSTEMS AND BIODIVERSITY LOSS

Theme: ecosystems and biodiversity loss

What follows concerns ecosystems and biodiversity loss. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

8.1 Biodiversity and Ecosystem Services

Carl Folke (2010) puts it plainly: resilience thinking reframes crisis as reorganization, focusing on thresholds, diversity, and learning.

Folke's resilience thinking treats crises as reorganisation rather than collapse. Thresholds, diversity and learning determine whether a system absorbs a shock or flips into a less desirable state. Management therefore aims at keeping options open. For biodiversity and ecosystem services, the practical consequence is to name the decision that would have to change.

The record on biodiversity and ecosystem services is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~800 million people face food insecurity (FAO, 2023). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how conservation and land management should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within conservation and land management, the test is whether the arrangement survives contact with a bad week.

How a problem is framed decides what counts as a solution, and biodiversity and ecosystem services is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as conservation and land management is concerned, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: Arctic sea ice declines by ~13% per decade (NASA, 2024). In the work of conservation and land management, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Uncertainty in biodiversity and ecosystem services is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~800 million people face food insecurity (FAO, 2023). For conservation and land management, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The news cycle supplies a concrete marker: Coral bleaching events are becoming more frequent, threatening marine ecosystems (IPCC, NOAA). Set beside the record, the episode indicates direction without fixing the rate. Within conservation and land management, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within conservation and land management, the operational reading is to record which limit applies to each decision before the decision is taken.

8.2 Habitat Loss

As Elizabeth Kolbert (2006) argues, extinction and climate change are not separate stories but one story of human planetary impact.

Kolbert's reporting treats extinction and climate change as one story of cumulative human pressure. Species loss removes the ecological margin that absorbs variability. The narrative connects field observation with paleoclimate evidence. Applied to habitat loss, the argument sets a test: does the measure still work when capacity is stretched?

It helps to ask what question habitat loss is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as conservation and land management is concerned, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: Arctic sea ice declines by ~13% per decade (NASA, 2024). In the work of conservation and land management, the practical implication is to plan for combinations

rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

An honest account of habitat loss distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~800 million people face food insecurity (FAO, 2023). Numbers carry confidence intervals, and those intervals should travel with them into any decision. For conservation and land management, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Evidence on habitat loss rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within conservation and land management, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

Current reporting adds detail rather than a new direction: Coral bleaching events are becoming more frequent, threatening marine ecosystems (IPCC, NOAA). Read against the longer record, it is an illustration rather than a proof. As far as conservation and land management is concerned, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as conservation and land management is concerned, the question is which of these decisions can be taken before the next event rather than during it.

8.3 Species Extinction Risk

As Johan Rockström (2009) argues, we define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.

Rockström's statement originates from the 2009 Nature paper 'A safe operating space for humanity' and its 2023 update that now finds six of nine boundaries transgressed. 'Planetary boundaries' are quantitative thresholds for climate, biosphere, freshwater, nutrients, and other Earth-system processes that define a Holocene-like stability zone. 'Operate safely' means staying within these thresholds reduces risk of abrupt, irreversible shifts. 'Six of nine transgressed' indicates humanity has left the safe zone on climate, biodiversity, land, freshwater, biogeochemical flows, and novel entities, far outside preindustrial variability. The update emphasizes interactions — transgressing one boundary tightens others, such as deforestation amplifying carbon release. The concept reframes species extinction risk from isolated sectoral problem to

systemic interdependence. For ecosystems and biodiversity loss, it implies solutions must be integrated across energy, food, and ecosystem management rather than siloed. For species extinction risk, the bearing is direct: the constraint is institutional as much as physical.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: Arctic sea ice declines by ~13% per decade (NASA, 2024). In the work of conservation and land management, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Not all statements about species extinction risk carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~800 million people face food insecurity (FAO, 2023). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. For conservation and land management, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The evidence base behind species extinction risk is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within conservation and land management, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Most disagreements about species extinction risk are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: Arctic sea ice declines by ~13% per decade (NASA, 2024). As far as conservation and land management is concerned, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

A recent marker points the same way: Coral bleaching events are becoming more frequent, threatening marine ecosystems (IPCC, NOAA). The value of such reports is that they date the change rather than explain it. In the work of conservation and land management, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of conservation and land management, the operational reading is to record which limit applies to each decision before the decision is taken.

8.4 Forest Ecosystems

Carl Folke (2010) puts it plainly: resilience thinking reframes crisis as reorganization, focusing on thresholds, diversity, and learning.

Folke's resilience thinking treats crises as reorganisation rather than collapse. Thresholds, diversity and learning determine whether a system absorbs a shock or flips into a less desirable state. Management therefore aims at keeping options open. Read against forest ecosystems, the point is that intention without capacity changes little.

The strength of the claim about forest ecosystems varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~800 million people face food insecurity (FAO, 2023). A central estimate without its range is not more precise, only less honest. For conservation and land management, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

The evidence base for forest ecosystems is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within conservation and land management, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

Discussion of forest ecosystems usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: Arctic sea ice declines by ~13% per decade (NASA, 2024). Read that way, the question for conservation and land management is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. As far as conservation and land management is concerned, the practical question is what would have to change first.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~800 million people face food insecurity (FAO, 2023). In the work of conservation and land management, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The news cycle supplies a concrete marker: Coral bleaching events are becoming more frequent, threatening marine ecosystems (IPCC, NOAA). It matters less as a record than as evidence that the baseline has already moved. For conservation and land management, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For conservation and land management, the question is which of these decisions can be taken before the next event rather than during it.

8.5 Case Study: Ecosystems and Biodiversity Loss

A single episode illustrates the mechanism: Coral bleaching events are becoming more frequent, threatening marine ecosystems (IPCC, NOAA). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. For conservation and land management, the episode fixes the date at which the argument stops being hypothetical.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which conservation and land management controls.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

8.6 Future Scenarios and Predictions: Ecosystems and Biodiversity Loss

For ecosystems and biodiversity, the near term is measured in bleaching and range shifts: by 2040 coral reefs face annual bleaching across 90% of tropical reefs, 15–25% of bird ranges move upslope, and mangroves lose 10–20% where seas rise fastest.

By 2050, one in ten species faces a high risk of extinction at 2 °C, and by 2100 narrow-range species face 2–3 times the extinction risk under high emissions; ecosystem services that shield coasts and pollinate crops weaken in step (IPCC, 2022; IPBES, 2019).

Marine protected networks, corridors linking forest patches, replanting diverse native species, and community ranger patrols keep ecosystems connected enough to adapt.

Table 8.1 Future scenarios and predictions for ecosystems and biodiversity loss (2030-2100)

Horizon	What changes for this chapter
2030	bleaching hits 50–70% of reefs
2050	1 in 10 species at high extinction risk
2100	narrow-range species risk ×2–3

Watch coral bleaching alerts, the arrival dates of migratory birds, and mangrove cover in your district; global reef and range trends are high-confidence, but local species losses vary with habitat fragmentation.

CHAPTER 9

HEALTH IMPACTS OF CLIMATE CHANGE

Theme: health impacts of climate change

This chapter examines health impacts of climate change. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

9.1 Direct and Indirect Health Impacts

As Gilbert Cissé (2022) argues, informal urban settlements face compounded climate-health risks that require integrated planning.

Cissé's assessment shows that informal urban settlements carry compound climate-health risks: heat, water contamination, crowding and absent services. Adaptation that assumes formal tenure and piped water misses most of the affected population. Planning has to start from the settlement, not the grid. In the terms of direct and indirect health impacts, the claim shifts attention from prediction to preparation.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~7 million air-pollution deaths occur annually (WHO, 2023). For that reason, average-based targets can be met while the most exposed end up worse off. For health systems and epidemiology, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. Within health systems and epidemiology, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. As far as health systems and epidemiology is concerned, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~7 million air-pollution deaths occur annually (WHO, 2023). None of those signals is a perfect measure, but together they indicate direction. In the work of health systems and epidemiology, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. As far as health systems and epidemiology is concerned, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For health systems and epidemiology, the operational reading is to record which limit applies to each decision before the decision is taken.

9.2 Heat-Related Illness

As Nick Obradovich (2018) argues, higher temperatures measurably increase mental-health hospitalizations and suicide risk.

Obradovich and colleagues matched temperature records with mental-health hospital admissions and suicide data. Both rise with heat, and the effect persists beyond a single day. The mechanism runs through sleep, medication and social contact. For heat-related illness, the practical consequence is to name the decision that would have to change.

Institutional performance in heat-related illness depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within health systems and epidemiology, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. As far as health systems and epidemiology is

concerned, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Anyone deciding about heat-related illness is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of health systems and epidemiology, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. For health systems and epidemiology, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Current reporting adds detail rather than a new direction: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). On its own it proves little; placed against the trend it shows the direction of travel. In the work of health systems and epidemiology, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within health systems and epidemiology, the question is which of these decisions can be taken before the next event rather than during it.

9.3 Vector-Borne Diseases

Kristie Ebi (2018) puts it plainly: if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with

co-benefits for productivity and equity. For vector-borne diseases within health impacts of climate change, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. Applied to vector-borne diseases, the argument sets a test: does the measure still work when capacity is stretched?

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. As far as health systems and epidemiology is concerned, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for vector-borne diseases is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of health systems and epidemiology, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~100 million people are affected by climate disasters annually (IFRC, 2022). Averages describe the size of the problem, not who carries it. Decisions taken by health systems and epidemiology — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. For health systems and epidemiology, the practical question is what would have to change first.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within health systems and epidemiology, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

A recent marker points the same way: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Taken alone the episode is weather; taken with the record it is a data point on a slope. For health systems and epidemiology, the report is a reminder that the response is being designed against a target that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as health systems and epidemiology is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

9.4 Water-Borne Diseases

As Caroline Hickman (2021) argues, 45% of youth say climate feelings impair daily life, tied to perceived government inaction.

Hickman's international survey of young people found widespread distress about climate and government inaction. Respondents reported that worry interfered with daily life and decisions about the future. The sample is large but self-selected in some countries. For water-borne diseases, the bearing is direct: the constraint is institutional as much as physical.

Any assessment of water-borne diseases will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~7 million air-pollution deaths occur annually (WHO, 2023). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of health systems and epidemiology, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in water-borne diseases the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~100 million people are affected by climate disasters annually (IFRC, 2022). For health systems and epidemiology, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What the record does show is that authority, money and information have to move together; any one without the others stalls. Within health systems and epidemiology, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The economics of water-borne diseases are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as health systems and epidemiology is concerned, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

The news cycle supplies a concrete marker: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). The item is useful because it fixes a date to a trend that otherwise drifts. Within health systems and epidemiology, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of health systems and epidemiology, the question is which of these decisions can be taken before the next event rather than during it.

9.5 Case Study: Health Impacts of Climate Change

The case that anchors this chapter is documented and recent: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. Within health systems and epidemiology, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). As far as health systems and epidemiology is concerned, the episode works as a readiness test rather than a weather story.

In the work of health systems and epidemiology, the transferable lesson is procedural: identify which decision failed first, and fix that decision rather than the symptom.

9.6 Future Scenarios and Predictions: Health Impacts of Climate Change

For health impacts, the near term shows up in clinics: by 2040 heat deaths rise 30–50% in cities above 35 °C, dengue expands 150–200 km poleward, and diarrheal cases jump 10–20% after floods.

By 2050, heat and vector-borne disease stretch health systems thin, with 500,000+ heat deaths a year already recorded; by 2100 highland malaria doubles under high emissions without control, and mental-health burden grows alongside physical

illness (Watts et al., 2021; WHO, 2023).

Heat-health alerts, screened housing, oral rehydration stocks, and clinics with flood-proof power and records protect the most exposed first.

Table 9.1 Future scenarios and predictions for health impacts of climate change (2030-2100)

Horizon	What changes for this chapter
2030	heat deaths +20–30%; dengue +100 km
2050	500k+ heat deaths/yr; systems strained
2100	highland malaria ×2; mental burden up

Watch heat-related admissions, dengue case counts, and clinic closures after storms; the direction of health risk is high-confidence, but local caseloads depend on sanitation, housing and health-system capacity.

CHAPTER 10

ECONOMIC IMPACTS AND FINANCIAL RESILIENCE

Theme: economic impacts and financial preparedness under climate change

The subject of this chapter is economic impacts and financial preparedness under climate change. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

10.1 Damage Cost Estimates

As Fatih Birol (2023) argues, the energy crisis sped the transition, but fossil demand must fall far faster for 1.5°C.

Birol's World Energy Outlook tracks the gap between stated goals and investment flows. Crises have accelerated renewables while leaving fossil demand too high for 1.5°C. The series is a scenario exercise, not a forecast. The implication for damage cost estimates is a question of sequence: what has to be in place before the event, not after it.

The economics of damage cost estimates are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of finance ministries and insurers, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Most disagreements about damage cost estimates are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For finance ministries and insurers, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind damage cost estimates is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within finance ministries and insurers, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as finance ministries and insurers is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The news cycle supplies a concrete marker: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). Read against the longer record, it is an illustration rather than a proof. In the work of finance ministries and insurers, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. In the work of finance ministries and insurers, the operational reading is to record which limit applies to each decision before the decision is taken.

10.2 Sectoral Impacts

Andrew Dessler (2021) puts it plainly: greenhouse physics has been settled since the 1800s; debate is about impacts and choices, not reality.

Dessler points out that the physics of the greenhouse effect has been settled since the nineteenth century. What remains open is the magnitude of impacts and the choices that determine them. Public debate often confuses the two. In the terms of sectoral impacts, the claim shifts attention from prediction to preparation.

Discussion of sectoral impacts usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Read that way, the question for finance ministries and insurers is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For finance ministries and insurers, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for sectoral impacts is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). The appropriate response to a strong signal with weak

local detail is not to wait for precision but to design for the range. Within finance ministries and insurers, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as finance ministries and insurers is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. In the work of finance ministries and insurers, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

Current reporting adds detail rather than a new direction: US 2023 billion-dollar disasters hit record 28 events, costing $92.9B (NOAA NCEI). The value of such reports is that they date the change rather than explain it. For finance ministries and insurers, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. For finance ministries and insurers, the question is which of these decisions can be taken before the next event rather than during it.

10.3 Labor Productivity Loss

As Naomi Klein (2014) argues, climate change changes everything about how we live, consume, and govern.

Klein connects climate policy to the economic model that produced it. She argues that market instruments alone cannot deliver the pace required. The claim is a political argument rather than an empirical finding. For labor productivity loss, the practical consequence is to name the decision that would have to change.

The record on labor productivity loss is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What remains contested is

attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how finance ministries and insurers should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within finance ministries and insurers, the test is whether the arrangement survives contact with a bad week.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as finance ministries and insurers is concerned, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. In the work of finance ministries and insurers, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

How a problem is framed decides what counts as a solution, and labor productivity loss is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For finance ministries and insurers, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). It matters less as a record than as evidence that the baseline has already moved. Within finance ministries and insurers, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within finance ministries and insurers, the operational reading is to record which limit applies to each decision before the decision is taken.

10.4 Infrastructure Damage

As Marshall Burke (2015) argues, economic output peaks at cool temperatures and declines sharply with heat, with non-linear losses for warm countries.

Burke and colleagues found that economic output peaks at moderate temperatures and falls sharply in hot conditions. Losses are non-linear and concentrated in tropical and already-warm economies. The result implies that warming redistributes income as much as it destroys it. Applied to infrastructure damage, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as finance ministries and insurers is concerned, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. In the work of finance ministries and insurers, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

It helps to ask what question infrastructure damage is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For finance ministries and insurers, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on infrastructure damage rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~100 million people are affected by climate disasters annually (IFRC, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within finance ministries and insurers, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: US 2023 billion-dollar disasters hit record 28 events, costing $92.9B (NOAA NCEI). Treated as a single observation it invites over-reading; treated as one of many it is informative. As far as finance

ministries and insurers is concerned, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as finance ministries and insurers is concerned, the question is which of these decisions can be taken before the next event rather than during it.

10.5 Case Study: Economic Impacts and Financial Resilience

One documented episode makes the argument concrete: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. As far as finance ministries and insurers is concerned, the record supplies both the warning and the timetable.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). In the work of finance ministries and insurers, the episode works as a readiness test rather than a weather story.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

10.6 Future Scenarios and Predictions: Economic Impacts and Financial Resilience

For economic impacts and financial resilience, the near term is visible in budgets: by 2040 climate disasters cost 2–4% of GDP a year in exposed provinces, coastal insurance premiums rise 30–60%, and heat cuts work hours 10–15%.

By 2050, loss and damage claims and adaptation costs together reach hundreds of billions a year, and by 2100 a high-emissions path shaves 15–25% off incomes in vulnerable deltas compared with a low-emissions path (IPCC, 2022; World Bank, 2022).

Emergency funds equal to three months of expenses, micro-insurance pools, diversified incomes, and building codes for wind, flood and heat reduce the shock.

Table 10.1 Future scenarios and predictions for economic impacts and financial preparedness under climate change (2030-2100)

Horizon	What changes for this chapter
2030	disaster cost 1–2% GDP; premiums +20%
2050	hundreds of billions/yr; adaptation gap
2100	incomes −15–25% in vulnerable deltas

Watch local insurance renewals, disaster-related budget reallocations, and days of work lost to heat; global cost estimates are medium-confidence, but household exposure can be tracked directly.

CHAPTER 11

MIGRATION AND DISPLACEMENT

Theme: climate migration and displacement

This chapter deals with climate migration and displacement. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

11.1 Drivers of Climate Migration

Diana Liverman (2020) puts it plainly: climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For drivers of climate migration within climate migration and displacement, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. Read against drivers of climate migration, the point is that intention without capacity changes little.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~100 million people are affected by climate disasters annually (IFRC, 2022). Averages describe the size of the problem, not who carries it. Decisions taken by migration policy and urban services — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as migration policy and urban services is concerned, that is the difference between a plan that can be defended and one that merely exists.

An honest account of drivers of climate migration distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of migration policy and urban services, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For migration policy and urban services, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The relevant question for drivers of climate migration is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within migration policy and urban services, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The news cycle supplies a concrete marker: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). On its own it proves little; placed against the trend it shows the direction of travel. For migration policy and urban services, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. As far as migration policy and urban services is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

11.2 Internal Displacement

As Saleemul Huq (2020) argues, it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For internal displacement within climate migration and displacement, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. The implication for

internal displacement is a question of sequence: what has to be in place before the event, not after it.

Not all statements about internal displacement carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of migration policy and urban services, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

Institutional performance in internal displacement depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~20 million people are displaced by climate annually (UNHCR, 2023). For migration policy and urban services, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Any assessment of internal displacement will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. Within migration policy and urban services, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in internal displacement the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as migration policy and urban services is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Current reporting adds detail rather than a new direction: Mediterranean wildfires in Greece (July 2023) forced 20,000 evacuations on Rhodes, largest in Greek history (Greek Civil Protection). Taken alone the episode is weather; taken with the record it is a data point on a slope. Within migration policy and urban services, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. In the work of migration policy and urban services, the question is which of these decisions can be taken before the next event rather than during it.

11.3 Cross-Border Movement

As Diana Liverman (2020) argues, climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For cross-border movement within climate migration and displacement, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. In the terms of cross-border movement, the claim shifts attention from prediction to preparation.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~20 million people are displaced by climate annually (UNHCR, 2023). For migration policy and urban services, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~100 million people are affected by climate disasters annually (IFRC, 2022). None of those signals is a perfect measure, but together they indicate direction. Within migration policy and urban services, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~500,000+ heat-related deaths occur annually (Watts et

al., 2021). For that reason, average-based targets can be met while the most exposed end up worse off. As far as migration policy and urban services is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The strength of the claim about cross-border movement varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~20 million people are displaced by climate annually (UNHCR, 2023). A central estimate without its range is not more precise, only less honest. In the work of migration policy and urban services, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

A recent marker points the same way: Amazon drought in late 2023 lowered Rio Negro to 120-year low, isolating 500,000 people (Brazil INPE, Reuters). The item is useful because it fixes a date to a trend that otherwise drifts. As far as migration policy and urban services is concerned, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For migration policy and urban services, the operational reading is to record which limit applies to each decision before the decision is taken.

11.4 Legal Status and Rights

Saleemul Huq (2020) puts it plainly: it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For legal status and rights within climate migration and displacement, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. For legal status and rights, the practical consequence is to name the decision that would have to change.

Anyone deciding about legal status and rights is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence

matters because early options buy the time that later options need. The pressure is documented rather than projected: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within migration policy and urban services, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as migration policy and urban services is concerned, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Uncertainty in legal status and rights is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~20 million people are displaced by climate annually (UNHCR, 2023). In the work of migration policy and urban services, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~100 million people are affected by climate disasters annually (IFRC, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. For migration policy and urban services, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: Libya's Derna dam collapse after Storm Daniel (Sept 2023) killed >4,300 and displaced 40,000 (UN OCHA, Libyan Red Crescent). What the report adds is timing, not a new mechanism. In the work of migration policy and urban services, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within migration policy and urban services, the question is which of these decisions can be taken before the next event rather than during it.

11.5 Case Study: Migration and Displacement

A single episode illustrates the mechanism: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. In the work of migration policy and urban services, the sequence is worth reconstructing before the next season rather than after it.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). For migration policy and urban services, the episode works as a readiness test rather than a weather story.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

11.6 Future Scenarios and Predictions: Migration and Displacement

For migration and displacement, the near term is counted in people on the move: by 2040, 15–25 million a year are displaced by floods, storms and drought, and cities such as Dhaka, Lagos and Manila absorb 300,000–500,000 climate newcomers annually.

By 2050, slow-onset loss of farmland and coastal land adds to sudden displacement, and by 2100 a high-emissions path displaces 100–150 million cumulatively from sub-3 m coasts and drying farmland (UNHCR, 2023; IPCC, 2022).

Legal aid, portable IDs and health records, rental support, and receiving-city plans for water, jobs and schools turn displacement from crisis into managed transition.

Table 11.1 Future scenarios and predictions for climate migration and displacement (2030-2100)

Horizon	What changes for this chapter
2030	15–20 m displaced/yr; +300k city arrivals
2050	slow-onset adds to sudden moves
2100	100–150 m cumulative (high path)

Watch arrival numbers in your city, shelter occupancy, and school enrollment of newcomer children; displacement totals are medium-confidence, but local inflows are recorded and can guide planning.

CHAPTER 12

CONFLICT, SECURITY AND STABILITY

Theme: climate change, conflict and security

What follows concerns climate change, conflict and security. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

12.1 Resource Competition

As Solomon Hsiang (2014) argues, climate variability is robustly linked to conflict risk, though institutions decide whether stress becomes violence.

Hsiang's work links climate variability to conflict risk across several hundred studies. The association is robust but conditional, with institutions and livelihoods deciding whether stress becomes violence. Climate is a risk multiplier, not a cause. For resource competition, the bearing is direct: the constraint is institutional as much as physical.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. Within security services and local government, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

How a problem is framed decides what counts as a solution, and resource competition is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~20 million people are displaced by climate annually (UNHCR, 2023). As far as security services and local government is concerned, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on resource competition is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how security services and local government should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. In the work of

security services and local government, the implication is a shorter list of decisions than the analysis suggests.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For security services and local government, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

The news cycle supplies a concrete marker: Horn of Africa's 2022 failed rainy seasons pushed 22 million into acute food insecurity (FEWS NET, IPC). The value of such reports is that they date the change rather than explain it. Within security services and local government, what matters is not the record but the lead time it implies.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. Within security services and local government, the operational reading is to record which limit applies to each decision before the decision is taken.

12.2 Climate and Violence Evidence

As Solomon Hsiang (2014) argues, climate variability is robustly linked to conflict risk, though institutions decide whether stress becomes violence.

Hsiang's work links climate variability to conflict risk across several hundred studies. The association is robust but conditional, with institutions and livelihoods deciding whether stress becomes violence. Climate is a risk multiplier, not a cause. Read against climate and violence evidence, the point is that intention without capacity changes little.

It helps to ask what question climate and violence evidence is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~20 million people are displaced by climate annually (UNHCR, 2023). As far as security services and local government is concerned, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on climate and violence evidence rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~100 million people are affected by climate disasters annually (IFRC, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. In the

work of security services and local government, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For security services and local government, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. Within security services and local government, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

Current reporting adds detail rather than a new direction: Brazil's Rio Grande do Sul floods (Apr-May 2024) displaced 600,000, deadliest climate disaster in state history (Brazil Civil Defense). It matters less as a record than as evidence that the baseline has already moved. As far as security services and local government is concerned, the item is useful mainly as a date stamp on a trend.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. As far as security services and local government is concerned, the question is which of these decisions can be taken before the next event rather than during it.

12.3 Fragile States

Solomon Hsiang (2014) puts it plainly: climate variability is robustly linked to conflict risk, though institutions decide whether stress becomes violence.

Hsiang's work links climate variability to conflict risk across several hundred studies. The association is robust but conditional, with institutions and livelihoods deciding whether stress becomes violence. Climate is a risk multiplier, not a cause. The implication for fragile states is a question of sequence: what has to be in place before the event, not after it.

The evidence base behind fragile states is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place

(IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of security services and local government, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For security services and local government, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Institutional performance in fragile states depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~20 million people are displaced by climate annually (UNHCR, 2023). Within security services and local government, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Most disagreements about fragile states are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as security services and local government is concerned, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

A recent marker points the same way: Cyclone Mocha (May 2023) displaced 1.7 million in Myanmar and Bangladesh (UN OCHA, WMO). Treated as a single observation it invites over-reading; treated as one of many it is informative. In the work of security services and local government, the report is a reminder that the response is being designed against a target that keeps moving.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. In the work of security services and local government, the operational reading is to record which limit applies to each decision before the decision

is taken.

12.4 Food and Water Conflict

As Peter Gleick (2020) argues, water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. In the terms of food and water conflict, the claim shifts attention from prediction to preparation.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For security services and local government, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~20 million people are displaced by climate annually (UNHCR, 2023). Within security services and local government, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

Discussion of food and water conflict usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~100 million people are affected by climate disasters annually (IFRC, 2022). Read that way, the question for security services and local government is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. As far as security services and local government is concerned, the practical question is what would have to change first.

The evidence base for food and water conflict is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. In the work of security services and local government, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before

acting are plans that act late.

The news cycle supplies a concrete marker: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). It is one line in a longer ledger, and the ledger is what matters. For security services and local government, what matters is not the record but the lead time it implies.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. For security services and local government, the question is which of these decisions can be taken before the next event rather than during it.

12.5 Case Study: Conflict, Security and Stability

The case that anchors this chapter is documented and recent: Horn of Africa's 2022 failed rainy seasons pushed 22 million into acute food insecurity (FEWS NET, IPC). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. For security services and local government, the episode fixes the date at which the argument stops being hypothetical.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence security services and local government can measure in its own records.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

12.6 Future Scenarios and Predictions: Conflict, Security and Stability

For conflict and security, the near term appears where rains fail repeatedly: by 2040 competition over water and grazing drives 20–30% more local disputes in the Sahel and Horn when drought lasts 2–3 years, and storm-driven outages add to civil unrest.

By 2050, food-price spikes and displacement pressure fragile states, and by 2100 a high-emissions path widens security gaps where institutions are already weak (IPCC, 2022; Mach et al., 2019).

Community water-peace committees, early-warning SMS, shared stockpiles, and transparent governance that shares resources before scarcity peaks reduce the trigger.

Table 12.1 Future scenarios and predictions for climate change, conflict and security (2030-2100)

Horizon	What changes for this chapter
2030	local disputes +10–15% in dry years
2050	food-price spikes; fragile states stressed
2100	security gaps widen (high path)

Watch local water-point conflicts, food-price protests, and displacement inflows across borders; climate's link to conflict is context-dependent and medium-confidence, strongest where governance is weak.

CHAPTER 13

INFRASTRUCTURE, SHELTER AND ENERGY RESILIENCE

Theme: infrastructure, shelter and energy resilience

This chapter examines infrastructure, shelter and energy resilience. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

13.1 Critical Infrastructure and Power Grids

As Raj Patel (2022) argues, hunger reflects power, not scarcity — climate shocks deepen those power gaps.

Patel argues that hunger reflects power and access rather than global scarcity. Climate shocks deepen existing gaps in entitlement. The framing shifts attention from production to distribution. Applied to critical infrastructure and power grids, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For infrastructure operators and utilities, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The strength of the claim about critical infrastructure and power grids varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. Within infrastructure operators and utilities, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. As far as infrastructure operators and utilities is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. In the work of infrastructure operators and utilities, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). Taken alone the episode is weather; taken with the record it is a data point on a slope. As far as infrastructure operators and utilities is concerned, the item is useful mainly as a date stamp on a trend.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. For infrastructure operators and utilities, the operational reading is to record which limit applies to each decision before the decision is taken.

13.2 Water and Sanitation Systems

Peter Gleick (2020) puts it plainly: water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. For water and sanitation systems, the bearing is direct: the constraint is institutional as much as physical.

Uncertainty in water and sanitation systems is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within infrastructure operators and utilities, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as infrastructure operators and utilities is concerned, equity is therefore a design criterion rather than a footnote. Measures that

reach the most exposed first tend to perform better for everyone else as well.

The economics of water and sanitation systems are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). In the work of infrastructure operators and utilities, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

Current reporting adds detail rather than a new direction: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). The item is useful because it fixes a date to a trend that otherwise drifts. In the work of infrastructure operators and utilities, the report is a reminder that the response is being designed against a target that keeps moving.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. Within infrastructure operators and utilities, the question is which of these decisions can be taken before the next event rather than during it.

13.3 Transport and Communication Networks

As Kevin Anderson (2020) argues, no new fossil exploration fits a 1.5°C budget — it is already spent.

Anderson's carbon-budget arithmetic implies that no new fossil extraction is compatible with 1.5°C. The budget is small enough that existing infrastructure already exhausts it. The claim is arithmetical and depends on the chosen probability of overshoot. Read against transport and communication networks, the point is that intention without capacity changes little.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Averages describe the size of the problem, not who carries it.

Decisions taken by infrastructure operators and utilities — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as infrastructure operators and utilities is concerned, that is the difference between a plan that can be defended and one that merely exists.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. In the work of infrastructure operators and utilities, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

An honest account of transport and communication networks distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. Within infrastructure operators and utilities, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

A recent marker points the same way: California's atmospheric rivers in winter 2022-23 ended drought but caused $4.6B flood damages (NOAA, CA DWR). What the report adds is timing, not a new mechanism. For infrastructure operators and utilities, what matters is not the record but the lead time it implies.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. As far as infrastructure operators and utilities is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

13.4 Building Standards and Cascading Failures

As Francesco La Camera (2023) argues, renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. The implication for building standards and cascading failures is a question of sequence: what has to be in place before the event, not after it.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. In the work of infrastructure operators and utilities, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Not all statements about building standards and cascading failures carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. Within infrastructure operators and utilities, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The aggregate picture hides the distribution, and in building standards and cascading failures the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). As far as infrastructure operators and utilities is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). The event is best read as a measurement, not as a

verdict. Within infrastructure operators and utilities, the item is useful mainly as a date stamp on a trend.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. In the work of infrastructure operators and utilities, the question is which of these decisions can be taken before the next event rather than during it.

13.5 Case Study: Infrastructure, Shelter and Energy Resilience

One documented episode makes the argument concrete: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. Within infrastructure operators and utilities, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence infrastructure operators and utilities can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

13.6 Future Scenarios and Predictions: Infrastructure, Shelter and Energy Resilience

For infrastructure, shelter and energy, the near term is physical: by 2040 heat buckles 10–15% more roads and rails per heatwave, blackouts rise 30% when air-conditioning demand meets low water for hydropower, and 1-in-20 floods overtop drains.

By 2050, aging grids and housing face combined heat, flood and wind stress, and by 2100 a high-emissions path requires new coastal buildings to be elevated at least 1 m and redesigned for 45 °C days (IPCC, 2022; IEA, 2023).

Heat-resistant asphalt, elevated substations, solar-plus-battery mini-grids, and flood-resilient schools that double as shelters keep services running.

Table 13.1 Future scenarios and predictions for infrastructure, shelter and energy resilience (2030-2100)

Horizon	What changes for this chapter
2030	outages +20%; road damage +10%
2050	combined heat-flood-wind stress
2100	+1 m elevation; 45 °C design days

Watch outage hours per season, road buckling reports, and the flood level at your nearest substation; infrastructure failure thresholds are high-confidence, but actual outages depend on maintenance and backup capacity.

CHAPTER 14

CLIMATE LITERACY AND EDUCATION

Theme: climate literacy and education

The subject of this chapter is climate literacy and education. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

14.1 What Is Climate Literacy

John Cook (2020) puts it plainly: misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. For the meaning of climate literacy, the practical consequence is to name the decision that would have to change.

How a problem is framed decides what counts as a solution, and the meaning of climate literacy is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of schools and adult education, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

Any assessment of the meaning of climate literacy will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. For schools and adult education, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The evidence base behind the meaning of climate literacy is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within schools and adult education, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it

is decoration.

Not all statements about the meaning of climate literacy carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~100 million people are affected by climate disasters annually (IFRC, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. As far as schools and adult education is concerned, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The news cycle supplies a concrete marker: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). It matters less as a record than as evidence that the baseline has already moved. In the work of schools and adult education, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of schools and adult education, the operational reading is to record which limit applies to each decision before the decision is taken.

14.2 Core Concepts and Age-Appropriate Education

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For core concepts and age-appropriate education within climate literacy and education, her message directs the analysis toward actionable, locally resonant communication and education strategies. Applied to core concepts and age-appropriate education, the argument sets a test: does the measure still work when capacity is stretched?

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already

shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. For schools and adult education, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The evidence base for core concepts and age-appropriate education is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within schools and adult education, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The strength of the claim about core concepts and age-appropriate education varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. As far as schools and adult education is concerned, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question core concepts and age-appropriate education is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of schools and adult education, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Current reporting adds detail rather than a new direction: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Treated as a single observation it invites over-reading; treated as one of many it is informative. For schools and adult education, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. For schools and adult education, the question is which of these decisions can be taken before the next event rather than during it.

14.3 Formal and Informal Education

As John Cook (2020) argues, misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. For formal and informal education, the bearing is direct: the constraint is institutional as much as physical.

The record on formal and informal education is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how schools and adult education should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within schools and adult education, the test is whether the arrangement survives contact with a bad week.

Uncertainty in formal and informal education is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as schools and adult education is concerned, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about formal and informal education are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of schools and adult education, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

Anyone deciding about formal and informal education is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For schools and adult education, the test of

a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

A recent marker points the same way: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). It is one line in a longer ledger, and the ledger is what matters. Within schools and adult education, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. Within schools and adult education, the operational reading is to record which limit applies to each decision before the decision is taken.

14.4 Teacher Training and Curriculum Design

Katharine Hayhoe (2014) puts it plainly: climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For teacher training and curriculum design within climate literacy and education, her message directs the analysis toward actionable, locally resonant communication and education strategies. Read against teacher training and curriculum design, the point is that intention without capacity changes little.

An honest account of teacher training and curriculum design distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~100 million people are affected by climate disasters annually (IFRC, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. As far as schools and adult education is concerned, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Discussion of teacher training and curriculum design usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: economic losses from climate disasters reach ~$200–300 billion per year (World Bank,

2022). Read that way, the question for schools and adult education is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. In the work of schools and adult education, the implication is a shorter list of decisions than the analysis suggests.

The relevant question for teacher training and curriculum design is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For schools and adult education, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

Evidence on teacher training and curriculum design rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~100 million people are affected by climate disasters annually (IFRC, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within schools and adult education, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Set beside the record, the episode indicates direction without fixing the rate. As far as schools and adult education is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. As far as schools and adult education is concerned, the question is which of these decisions can be taken before the next event rather than during it.

14.5 Case Study: Climate Literacy and Education

A single episode illustrates the mechanism: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. As far as schools and adult education is concerned, the record supplies both the warning and the timetable.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which schools and adult education controls.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

14.6 Future Scenarios and Predictions: Climate Literacy and Education

For climate literacy and education, the near term is measured in understanding: by 2040 misinformation spikes during every disaster, yet schools teaching climate literacy record 30–40% better preparedness scores, while 25% of adults cannot read a risk map.

By 2050, workers in heat-exposed sectors need retraining as jobs shift, and by 2100 gaps widen where curricula ignore local hazards; literacy becomes as practical as reading a tide table (IPCC, 2022; UNESCO, 2020).

Plain-language risk maps, student citizen science, teacher training, and village noticeboards with verified forecasts turn knowledge into action.

Table 14.1 Future scenarios and predictions for climate literacy and education (2030-2100)

Horizon	What changes for this chapter
2030	preparedness +30–40% in literate schools
2050	retraining for heat-exposed jobs
2100	gaps widen where curricula ignore hazards

Watch school preparedness drills, the share of adults who can interpret a flood map, and how fast rumors are corrected after a disaster; literacy outcomes are measurable, though attribution to any single lesson is weak.

CHAPTER 15

DATA, STATISTICS AND MISINFORMATION

Theme: data, statistics and misinformation in the climate era

This chapter deals with data, statistics and misinformation in the climate era. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

15.1 Why Data Matters and Types of Data

As John Cook (2020) argues, misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. In the terms of why data matter and which data are used, the claim shifts attention from prediction to preparation.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Averages describe the size of the problem, not who carries it. Decisions taken by statistical agencies and the media — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as statistical agencies and the media is concerned, that is the difference between a plan that can be defended and one that merely exists.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of statistical agencies and the media, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. For statistical agencies and the media, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for why data matter and which data are used is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance,

redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within statistical agencies and the media, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The news cycle supplies a concrete marker: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). The item is useful because it fixes a date to a trend that otherwise drifts. For statistical agencies and the media, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as statistical agencies and the media is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

15.2 Descriptive Statistics and Uncertainty

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For descriptive statistics and uncertainty within data, her message directs the analysis toward actionable, locally resonant communication and education strategies. For descriptive statistics and uncertainty, the practical consequence is to name the decision that would have to change.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~100 million people are affected by climate disasters annually (IFRC, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. In the work of statistical agencies and the media, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The economics of descriptive statistics and uncertainty are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For statistical agencies and the media, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Any assessment of descriptive statistics and uncertainty will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. Within statistical agencies and the media, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in descriptive statistics and uncertainty the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as statistical agencies and the media is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Current reporting adds detail rather than a new direction: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). What the report adds is timing, not a new mechanism. Within statistical agencies and the media, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. In the work of statistical agencies and the media, the question is which of these decisions can be taken before the next event rather than during it.

15.3 Correlation vs Causation

John Cook (2020) puts it plainly: misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. Applied to the difference between correlation and causation, the argument sets a test: does the measure still work when capacity is stretched?

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. For statistical agencies and the media, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. Within statistical agencies and the media, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~100 million people are affected by climate disasters annually (IFRC, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. As far as statistical agencies and the media is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. In the work of statistical agencies and the media, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

A recent marker points the same way: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). The event is best read as a measurement, not as a verdict. As far as statistical agencies and the media is concerned, the report is a reminder that the response is being designed against a target

that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. For statistical agencies and the media, the operational reading is to record which limit applies to each decision before the decision is taken.

15.4 Graphs, Charts and Misleading Statistics

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For graphs within data, her message directs the analysis toward actionable, locally resonant communication and education strategies. For graphs, the bearing is direct: the constraint is institutional as much as physical.

Anyone deciding about graphs, charts and misleading statistics is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within statistical agencies and the media, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as statistical agencies and the media is concerned, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in graphs, charts and misleading statistics depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). In the work of statistical agencies and the media, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. For statistical agencies and the media, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

The news cycle supplies a concrete marker: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. In the work of statistical agencies and the media, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. Within statistical agencies and the media, the question is which of these decisions can be taken before the next event rather than during it.

15.5 Case Study: Data, Statistics and Misinformation

The case that anchors this chapter is documented and recent: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. In the work of statistical agencies and the media, the sequence is worth reconstructing before the next season rather than after it.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which statistical agencies and the media controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

15.6 Future Scenarios and Predictions: Data, Statistics and Misinformation

For data, statistics and misinformation, the near term is a race between signals and noise: by 2040, 60–70% of online climate claims mix real and outdated data, and gaps between satellites and ground stations delay warnings by 6–12 hours.

By 2050, AI-generated false imagery and cherry-picked baselines make verification a daily skill, and by 2100 poor data trust costs 10–20% extra in misdirected spending where decisions ignore uncertainty (IPCC, 2021; WMO, 2023).

Check primary sources (IPCC, WMO, NOAA/NASA), triangulate two or three datasets, and report ranges with uncertainty bars rather than single numbers.

Table 15.1 Future scenarios and predictions for data, statistics and misinformation in the climate era (2030-2100)

Horizon	What changes for this chapter
2030	60% mixed claims; warnings delayed 6 h
2050	AI false imagery; daily verification
2100	10–20% wasted spending where trust fails

Watch whether a claim cites a baseline year, whether it matches the nearest ground station, and whether it survives a second source; data quality is checkable, unlike vague predictions.

CHAPTER 16

HOUSEHOLD PREPAREDNESS: WATER, FOOD, SHELTER AND RISK

Theme: household preparedness for water, food, shelter and local risk

What follows concerns household preparedness for water, food, shelter and local risk. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

16.1 Local Hazard Mapping and Personal Risk Matrix

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For local hazard mapping and personal risk matrix within household preparedness for water, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. The implication for local hazard mapping and personal risk matrix is a question of sequence: what has to be in place before the event, not after it.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~2 billion people lack safe water (WHO & UNICEF, 2023). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within households and community groups, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

How a problem is framed decides what counts as a solution, and local hazard mapping and personal risk matrix is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). As far as households and community groups is concerned, the

practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on local hazard mapping and personal risk matrix is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how households and community groups should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. In the work of households and community groups, the implication is a shorter list of decisions than the analysis suggests.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The news cycle supplies a concrete marker: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). Treated as a single observation it invites over-reading; treated as one of many it is informative. Within households and community groups, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. Within households and community groups, the operational reading is to record which limit applies to each decision before the decision is taken.

16.2 Daily Water Needs, Storage and Purification

W. Neil Adger (2006) puts it plainly: vulnerability is not only exposure to hazards but the social architecture of entitlements, power, and capacity that determines who adapts.

Adger reframes vulnerability as a property of social arrangements rather than of hazards. Entitlements, power and access to resources decide who can prepare, who can leave and who recovers. The point is structural, not semantic. In the terms of daily water needs, the claim shifts attention from prediction to preparation.

It helps to ask what question daily water needs, storage and purification is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical,

because the second framing admits trade-offs that the first hides. The scale is already visible: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). As far as households and community groups is concerned, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on daily water needs, storage and purification rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. In the work of households and community groups, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within households and community groups, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

Current reporting adds detail rather than a new direction: India's wheat yields fell 10% in 2022 heatwave, prompting export ban affecting global prices (FAO, India IMD). It is one line in a longer ledger, and the ledger is what matters. As far as households and community groups is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as households and community groups is concerned, the question is which of these decisions can be taken before the next event rather than during it.

16.3 Rainwater Harvesting and Greywater Reuse

As David Lobell (2011) argues, warming since 1980 has already cut wheat and maize yields, masking gains from technology.

Lobell's analysis separated climate effects from technology and management in global crop records. Warming since 1980 reduced wheat and maize yields relative to a counterfactual without it. Adaptation and carbon fertilisation offset part, but not all, of the loss. For rainwater harvesting and greywater reuse, the practical consequence is to name the decision that would have to change.

The evidence base behind rainwater harvesting and greywater reuse is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of households and community groups, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

The economics of rainwater harvesting and greywater reuse are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within households and community groups, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Most disagreements about rainwater harvesting and greywater reuse are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as households and community groups is concerned, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in

advance is what separates analysis from advocacy.

A recent marker points the same way: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Set beside the record, the episode indicates direction without fixing the rate. In the work of households and community groups, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of households and community groups, the operational reading is to record which limit applies to each decision before the decision is taken.

16.4 Nutritional Needs and Food Storage

As Chuang Zhao (2017) argues, each degree of warming lowers staple yields by ~6%, with wheat most vulnerable to heat.

Zhao and colleagues combined field warming experiments with models to estimate yield responses per degree. Staples fall by roughly six per cent per degree of local warming, with wheat most sensitive. The estimate applies to present-day locations and practices. Applied to nutritional needs and food storage, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within households and community groups, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

Discussion of nutritional needs and food storage usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Read that way, the question for households and community groups is not whether the trend is real but

which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. As far as households and community groups is concerned, the practical question is what would have to change first.

The evidence base for nutritional needs and food storage is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~2 billion people lack safe water (WHO & UNICEF, 2023). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. In the work of households and community groups, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The news cycle supplies a concrete marker: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). Read against the longer record, it is an illustration rather than a proof. For households and community groups, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. For households and community groups, the question is which of these decisions can be taken before the next event rather than during it.

16.5 Case Study: Household Preparedness — Water, Food, Shelter and Risk

One documented episode makes the argument concrete: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. For households and community groups, the episode fixes the date at which the argument stops being hypothetical.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence households and community groups can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

16.6 Future Scenarios and Predictions: Household Preparedness — Water, Food, Shelter and Risk

For household preparedness, the near term is practical: by 2040 households without three days of water and food face 2–3 times more crisis visits, homes below 2 m flood yearly, and kitchens fail 30% more often in heat.

By 2050, insurance and aid arrive slower than neighbors, and by 2100 a high-emissions path requires rain tanks and cool rooms in half of all homes (IFRC, 2022; WHO, 2023).

Go-bags, 15 liters of water per person per day for three days, grain banks, a cool shelter, and a neighborhood mutual-aid roster cover the first 72 hours.

Table 16.1 Future scenarios and predictions for household preparedness for water, food, shelter and local risk (2030-2100)

Horizon	What changes for this chapter
2030	3-day kit cuts losses 20–30%
2050	aid lags; neighbors first responders
2100	rain tanks/cool rooms in 50% of homes

Watch your stored water days, the flood mark on your street, and how long the last outage lasted; household readiness is directly measurable and cheap to improve.

CHAPTER 17

HEALTH, EMERGENCY RESPONSE AND SPECIAL POPULATIONS

Theme: health, emergency response and protecting vulnerable populations

This chapter examines health, emergency response and protecting vulnerable populations. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

17.1 First Aid Basics and Heat Illness

Diana Liverman (2020) puts it plainly: climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For first aid basics and heat illness within health, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. Read against first aid basics and heat illness, the point is that intention without capacity changes little.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~7 million air-pollution deaths occur annually (WHO, 2023). For emergency services and clinics, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The strength of the claim about first aid basics and heat illness varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). A central estimate without its range is not more precise, only less honest. Within emergency services and clinics, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For that reason, average-based targets can be met while the most exposed end up worse off. As far as emergency services and clinics is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~7 million air-pollution deaths occur annually (WHO, 2023). None of those signals is a perfect measure, but together they indicate direction. In the work of emergency services and clinics, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). What the report adds is timing, not a new mechanism. As far as emergency services and clinics is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For emergency services and clinics, the operational reading is to record which limit applies to each decision before the decision is taken.

17.2 Wound Care, Infection and Medication Management

As Saleemul Huq (2020) argues, it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For wound care within health, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. The implication for wound care is a question of sequence:

what has to be in place before the event, not after it.

Uncertainty in wound care, infection and medication management is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within emergency services and clinics, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as emergency services and clinics is concerned, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Anyone deciding about wound care, infection and medication management is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of emergency services and clinics, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. For emergency services and clinics, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

Current reporting adds detail rather than a new direction: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). The event is best read as a measurement, not as a verdict. In the work of emergency services and clinics, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within emergency services and clinics, the question is which of these decisions can be taken before the next event rather than during it.

17.3 Chronic Disease and Elderly Care

As Maria Neira (2021) argues, climate change is the single biggest health threat facing humanity — the burning of fossil fuels is killing us.

Neira's statement was made as Director of the WHO Department of Environment, Climate Change and Health around the 2021 COP26 Special Report. 'Single biggest health threat facing humanity' elevates climate from environmental to health emergency, citing heat, air pollution, food and water insecurity, and infectious disease. 'Burning of fossil fuels is killing us' directly links emissions source to mortality — 7 million annual deaths from air pollution, hundreds of thousands from heat, plus wider burdens. Neira argues health professionals are trusted messengers who can make climate personal and urgent. The WHO report she launched called for health systems to be climate-resilient and low-carbon, and for climate policy to be judged on health co-benefits. For chronic disease and elderly care within health, her framing justifies prioritizing prevention, health surveillance linked to climate monitoring, and equity in access to care. In the terms of chronic disease and elderly care, the claim shifts attention from prediction to preparation.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Averages describe the size of the problem, not who carries it. Decisions taken by emergency services and clinics — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as emergency services and clinics is concerned, that is the difference between a plan that can be defended and one that merely exists.

The relevant question for chronic disease and elderly care is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of emergency services and clinics, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt

(UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For emergency services and clinics, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

An honest account of chronic disease and elderly care distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Numbers carry confidence intervals, and those intervals should travel with them into any decision. Within emergency services and clinics, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

A recent marker points the same way: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. For emergency services and clinics, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as emergency services and clinics is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

17.4 Children, Disabilities and Special Needs

Kristie Ebi (2018) puts it plainly: if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For children within health, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For children, the practical consequence is to name the decision that would have to change.

Any assessment of children, disabilities and special needs will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~7 million air-pollution deaths occur annually (WHO, 2023). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of emergency services and clinics, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

Institutional performance in children, disabilities and special needs depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For emergency services and clinics, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Not all statements about children, disabilities and special needs carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. Within emergency services and clinics, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The aggregate picture hides the distribution, and in children, disabilities and special needs the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as emergency services and clinics is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

The news cycle supplies a concrete marker: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). On its own it proves little; placed against the trend it shows the direction of travel. Within emergency services and clinics, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still

open. In the work of emergency services and clinics, the question is which of these decisions can be taken before the next event rather than during it.

17.5 Case Study: Health, Emergency Response and Special Populations

A single episode illustrates the mechanism: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. Within emergency services and clinics, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence emergency services and clinics can measure in its own records.

In the work of emergency services and clinics, the transferable lesson is procedural: identify which decision failed first, and fix that decision rather than the symptom.

17.6 Future Scenarios and Predictions: Health, Emergency Response and Special Populations

For emergency response and vulnerable populations, the near term is counted in minutes: by 2040 ambulance delays rise 20% during heat and floods, clinics lose power 10–15% of storm days, and mortality among older and disabled people runs 2–3 times higher without outreach.

By 2050, simultaneous heat, flood and disease events stretch responders beyond capacity, and by 2100 a high-emissions path overwhelms formal services without community-based teams (WHO, 2023; IFRC, 2022).

Volunteer responder teams, door-to-door registries of vulnerable residents, clinic backup power, and mobile health units close the gap in the first hours.

Table 17.1 Future scenarios and predictions for health, emergency response and protecting vulnerable populations (2030-2100)

Horizon	What changes for this chapter
2030	ambulance delays +20%; outages 10–15%
2050	compound events exceed capacity
2100	formal services overwhelmed without volunteers

Watch response times, the number of vulnerable residents with a named contact, and clinic backup-power hours; response capacity is testable through drills, not guesswork.

CHAPTER 18

SOCIAL CAPITAL, MUTUAL AID AND GOVERNANCE

Theme: social capital, mutual aid and community governance

The subject of this chapter is social capital, mutual aid and community governance. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

18.1 What Is Social Capital and Trust

As Daniel P. Aldrich (2015) argues, social networks, especially bridging ties, are the strongest predictor of post-disaster survival and recovery.

Aldrich finds that social capital, and especially bridging ties across groups, predicts post-disaster survival better than physical defences. Networks deliver information, shelter and credit before agencies arrive. Bonding ties alone can also exclude outsiders. For the meaning of social capital and trust, the bearing is direct: the constraint is institutional as much as physical.

Uncertainty in the meaning of social capital and trust is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of civil society and local associations, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about the meaning of social capital and trust are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~7 million air-pollution deaths occur annually (WHO, 2023). For civil society and local associations, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind the meaning of social capital and trust is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within civil society and local associations, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not

evidence; it is decoration.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

The news cycle supplies a concrete marker: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). It is one line in a longer ledger, and the ledger is what matters. In the work of civil society and local associations, what matters is not the record but the lead time it implies.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of civil society and local associations, the operational reading is to record which limit applies to each decision before the decision is taken.

18.2 Network Mapping, Bridging and Bonding

As Naomi Klein (2014) argues, climate change changes everything about how we live, consume, and govern.

Klein connects climate policy to the economic model that produced it. She argues that market instruments alone cannot deliver the pace required. The claim is a political argument rather than an empirical finding. Read against network mapping, the point is that intention without capacity changes little.

Discussion of network mapping, bridging and bonding usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~7 million air-pollution deaths occur annually (WHO, 2023). Read that way, the question for civil society and local associations is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For civil society and local associations, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for network mapping, bridging and bonding is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~100 million people are affected by climate disasters annually (IFRC, 2022). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within

civil society and local associations, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

An honest account of network mapping, bridging and bonding distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~7 million air-pollution deaths occur annually (WHO, 2023). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of civil society and local associations, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Current reporting adds detail rather than a new direction: Amazon drought in late 2023 lowered Rio Negro to 120-year low, isolating 500,000 people (Brazil INPE, Reuters). Set beside the record, the episode indicates direction without fixing the rate. For civil society and local associations, the item is useful mainly as a date stamp on a trend.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For civil society and local associations, the question is which of these decisions can be taken before the next event rather than during it.

18.3 Community Leaders and Inclusive Participation

Susan L. Cutter (2008) puts it plainly: resilience is place-based — community traits explain recovery far better than hazard magnitude alone.

Cutter's place-based work shows that community characteristics explain recovery better than hazard size does. Her social-vulnerability mapping combines age, income, tenure and access to services into an index that local planners can use. The method is a screening tool, not a verdict. The implication for community leaders and inclusive participation is a question of sequence: what has to be in place before the event, not after it.

The record on community leaders and inclusive participation is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is

consistent across methods (IPCC, 2023). The headline figure is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how civil society and local associations should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within civil society and local associations, the test is whether the arrangement survives contact with a bad week.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Not all statements about community leaders and inclusive participation carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~7 million air-pollution deaths occur annually (WHO, 2023). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of civil society and local associations, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and community leaders and inclusive participation is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~100 million people are affected by climate disasters annually (IFRC, 2022). For civil society and local associations, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Libya's Derna dam collapse after Storm Daniel (Sept 2023) killed >4,300 and displaced 40,000 (UN OCHA, Libyan Red Crescent). Read against the longer record, it is an illustration rather than a proof. Within civil society and local associations, the report is a reminder that the response is being designed against a target that keeps moving.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they

are also the easiest to check. Within civil society and local associations, the operational reading is to record which limit applies to each decision before the decision is taken.

18.4 Principles of Mutual Aid and Starting a Group

As Fikret Berkes (2013) argues, linking Indigenous and scientific knowledge creates legitimate, effective adaptation that neither can achieve alone.

Berkes documents how indigenous and local knowledge systems support adaptation through fine-grained observation over generations. Linking that knowledge with scientific monitoring produces options neither can generate alone. The combination requires reciprocity, not extraction. In the terms of principles of mutual aid and starting a group, the claim shifts attention from prediction to preparation.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The strength of the claim about principles of mutual aid and starting a group varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~7 million air-pollution deaths occur annually (WHO, 2023). A central estimate without its range is not more precise, only less honest. In the work of civil society and local associations, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question principles of mutual aid and starting a group is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~100 million people are affected by climate disasters annually (IFRC, 2022). For civil society and local associations, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on principles of mutual aid and starting a group rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within civil society and local associations, that means planning against a range

rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people (WMO, UN OCHA). The value of such reports is that they date the change rather than explain it. As far as civil society and local associations is concerned, what matters is not the record but the lead time it implies.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as civil society and local associations is concerned, the question is which of these decisions can be taken before the next event rather than during it.

18.5 Case Study: Social Capital, Mutual Aid and Governance

The case that anchors this chapter is documented and recent: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. As far as civil society and local associations is concerned, the record supplies both the warning and the timetable.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence civil society and local associations can measure in its own records.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

18.6 Future Scenarios and Predictions: Social Capital, Mutual Aid and Governance

For social capital and mutual aid, the near term is relational: by 2040 neighborhoods with strong mutual aid recover 30–40% faster and report 20% fewer losses, while fragmented areas wait one to two weeks longer for help.

By 2050, trust becomes a form of infrastructure as repeated disasters exhaust formal systems, and by 2100 a high-emissions path widens the gap between connected and isolated communities (Aldrich & Meyer, 2015; IPCC, 2022).

Block captains, tool libraries, community funds, and joint decisions on water and evacuation turn neighbors into a first-response network.

Table 18.1 Future scenarios and predictions for social capital, mutual aid and community governance (2030-2100)

Horizon	What changes for this chapter
2030	recovery 30–40% faster with strong ties
2050	trust as infrastructure; repeated shocks
2100	connected vs isolated gap widens

Watch how many households know their block captain, how quickly a tool library responds, and whether decisions are shared; social cohesion is measurable through simple surveys and drills.

CHAPTER 19

RENEWABLE ENERGY AND APPROPRIATE TECHNOLOGIES

Theme: renewable energy and appropriate technologies for climate adaptation

This chapter deals with renewable energy and appropriate technologies for climate adaptation. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

19.1 Renewable Energy Systems and Solar/Battery

As Saleemul Huq (2020) argues, it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For renewable energy within renewable energy and appropriate technologies for climate adaptation, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. Applied to renewable energy, the argument sets a test: does the measure still work when capacity is stretched?

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. As far as energy planners and regulators is concerned, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~100 million people are affected by climate disasters annually (IFRC, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. In the work of energy planners and regulators, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For energy planners and regulators, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within energy planners and regulators, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The news cycle supplies a concrete marker: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). The event is best read as a measurement, not as a verdict. For energy planners and regulators, the item is useful mainly as a date stamp on a trend.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as energy planners and regulators is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

19.2 Water Technologies and Purification

Fatih Birol (2023) puts it plainly: the energy crisis sped the transition, but fossil demand must fall far faster for 1.5°C.

Birol's World Energy Outlook tracks the gap between stated goals and investment flows. Crises have accelerated renewables while leaving fossil demand too high for 1.5°C. The series is a scenario exercise, not a forecast. For water technologies and purification, the bearing is direct: the constraint is institutional as much as physical.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. In the work of energy planners and regulators, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in water technologies and purification depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). For energy planners and regulators, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within energy planners and regulators, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Anyone deciding about water technologies and purification is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as energy planners and regulators is concerned, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Current reporting adds detail rather than a new direction: Vanuatu's twin cyclones Judy and Kevin (Mar 2023) hit within 48 hours, affecting 80% of population (UNICEF, WMO). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. Within energy planners and regulators, the report is a reminder that the response is being designed against a target that keeps moving.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of energy planners and regulators, the question is which of these decisions can be taken before the next event rather than during it.

19.3 Agricultural Technologies and Adaptive Practices

As Francesco La Camera (2023) argues, renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. Read against agricultural technologies and adaptive practices, the point is that intention without capacity changes little.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). For energy planners and regulators, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within energy planners and regulators, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for agricultural technologies and adaptive practices is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as energy planners and regulators is concerned, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). Averages describe the size of the problem, not who carries it. Decisions taken by energy planners and regulators — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. In the work of energy planners and regulators, the test is whether the arrangement survives contact with a bad week.

A recent marker points the same way: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). On its own it proves little; placed against the trend it shows the direction of travel. As far as energy planners and regulators is concerned, what matters is not the record but the lead time it implies.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For energy planners and regulators, the operational reading is to record which limit applies to each decision before the decision is taken.

19.4 Early Warning Systems and AI for Climate

As Inger Andersen (2023) argues, adaptation finance gaps are widening, leaving the vulnerable to pay with lives.

UNEP's adaptation-gap reports compare finance needs with flows actually disbursed. The gap has widened every year the series has been published. Adaptation remains the smaller share of climate finance by a wide margin. The implication for early warning systems and AI for climate is a question of sequence: what has to be in place before the event, not after it.

The economics of early warning systems and AI for climate are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within energy planners and regulators, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Any assessment of early warning systems and AI for climate will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. As far as energy planners and regulators is concerned, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in early warning systems and AI for climate the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). In the work of energy planners and regulators, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. For energy planners and regulators, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: US 2023 billion-dollar disasters hit record 28 events, costing $92.9B (NOAA NCEI). Taken alone the episode is weather; taken with the record it is a data point on a slope. In the work of energy planners and regulators, the item is useful mainly as a date stamp on a trend.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within energy planners and regulators, the question is which of these decisions can be taken before the next event rather than during it.

19.5 Case Study: Renewable Energy and Appropriate Technologies

One documented episode makes the argument concrete: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. In the work of energy planners and regulators, the sequence is worth reconstructing before the next season rather than after it.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence energy planners and regulators can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

19.6 Future Scenarios and Predictions: Renewable Energy and Appropriate Technologies

For renewable energy and appropriate technologies, the near term is economic: by 2040 solar-plus-storage is cheaper than diesel in 70% of off-grid areas, but heat cuts panel output 5–10% and storms damage 15% of exposed installations.

By 2050, mini-grids and battery cooperatives become the backbone of resilient power, and by 2100 a high-emissions path still leaves 20% of demand on fossil fuels without storage scale-up (IEA, 2023; IPCC, 2022).

Elevated solar arrays, cyclone clips, battery cooperatives, and repair training for young people keep power on when the grid fails.

Table 19.1 Future scenarios and predictions for renewable energy and appropriate technologies for climate adaptation (2030-2100)

Horizon	What changes for this chapter
2030	solar+storage cheaper in 50% off-grid
2050	mini-grids backbone; battery co-ops
2100	20% fossil demand without storage scale

Watch your system's output on the hottest day, battery autonomy hours, and the number of trained local technicians; technology costs are high-confidence, but storm resilience depends on installation quality.

CHAPTER 20

CLIMATE PSYCHOLOGY AND MENTAL HEALTH

Theme: climate anxiety, trauma and building psychological resilience

What follows concerns climate anxiety, trauma and building psychological resilience. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

20.1 Climate Anxiety and Rational Alarm

Kristie Ebi (2018) puts it plainly: if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For climate anxiety and rational alarm within climate anxiety, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For climate anxiety and rational alarm, the practical consequence is to name the decision that would have to change.

Most disagreements about climate anxiety and rational alarm are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within mental-health services and schools, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind climate anxiety and rational alarm is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as mental-health services and schools is concerned, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not

evidence; it is decoration.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Uncertainty in climate anxiety and rational alarm is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). For mental-health services and schools, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The news cycle supplies a concrete marker: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Set beside the record, the episode indicates direction without fixing the rate. Within mental-health services and schools, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within mental-health services and schools, the operational reading is to record which limit applies to each decision before the decision is taken.

20.2 Youth Anxiety and Government Inaction

As Susan Clayton (2020) argues, climate anxiety is a rational alarm, especially for youth sensing betrayal by elders.

Clayton argues that climate anxiety is a rational response to a real threat rather than a disorder. Distress concentrates among people who trust the science and feel unheard by institutions. Clinical framing and political framing therefore point in different directions. Applied to youth anxiety and government inaction, the argument sets a test: does the measure still work when capacity is stretched?

The evidence base for youth anxiety and government inaction is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~7 million air-pollution deaths occur annually (WHO, 2023). The appropriate response to a strong signal with weak

local detail is not to wait for precision but to design for the range. As far as mental-health services and schools is concerned, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

An honest account of youth anxiety and government inaction distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~100 million people are affected by climate disasters annually (IFRC, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. For mental-health services and schools, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Discussion of youth anxiety and government inaction usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~7 million air-pollution deaths occur annually (WHO, 2023). Read that way, the question for mental-health services and schools is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. Within mental-health services and schools, the test is whether the arrangement survives contact with a bad week.

Current reporting adds detail rather than a new direction: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Read against the longer record, it is an illustration rather than a proof. As far as mental-health services and schools is concerned, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as mental-health services and schools is concerned, the question is which of these decisions can be taken before the next event rather than during it.

20.3 Trauma, PTSD and Mental Health Impacts

As Kristie Ebi (2018) argues, if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For trauma within climate anxiety, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For trauma, the bearing is direct: the constraint is institutional as much as physical.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Not all statements about trauma, PTSD and mental health impacts carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~100 million people are affected by climate disasters annually (IFRC, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. For mental-health services and schools, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and trauma, PTSD and mental health impacts is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~7 million air-pollution deaths occur annually (WHO, 2023). Within mental-health services and schools, the practical consequence is that the frame

has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on trauma, PTSD and mental health impacts is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how mental-health services and schools should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. As far as mental-health services and schools is concerned, the practical question is what would have to change first.

A recent marker points the same way: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). The value of such reports is that they date the change rather than explain it. In the work of mental-health services and schools, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of mental-health services and schools, the operational reading is to record which limit applies to each decision before the decision is taken.

20.4 Building Psychological Resilience

Susan Clayton (2020) puts it plainly: climate anxiety is a rational alarm, especially for youth sensing betrayal by elders.

Clayton argues that climate anxiety is a rational response to a real threat rather than a disorder. Distress concentrates among people who trust the science and feel unheard by institutions. Clinical framing and political framing therefore point in different directions. Read against building psychological resilience, the point is that intention without capacity changes little.

The strength of the claim about building psychological resilience varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. For mental-health services and schools, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question building psychological resilience is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical,

because the second framing admits trade-offs that the first hides. The scale is already visible: ~7 million air-pollution deaths occur annually (WHO, 2023). Within mental-health services and schools, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on building psychological resilience rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. As far as mental-health services and schools is concerned, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of mental-health services and schools, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The news cycle supplies a concrete marker: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). It matters less as a record than as evidence that the baseline has already moved. For mental-health services and schools, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For mental-health services and schools, the question is which of these decisions can be taken before the next event rather than during it.

20.5 Case Study: Climate Psychology and Mental Health

A single episode illustrates the mechanism: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. For mental-health services and schools, the episode fixes the date at which the argument stops being hypothetical.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). Within mental-health services and schools, the episode works as a readiness test rather than a weather story.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

20.6 Future Scenarios and Predictions: Climate Psychology and Mental Health

For climate anxiety and mental health, the near term is emotional: by 2040, 35–45% of teenagers report weekly climate anxiety, heat and displacement raise depression 15–25%, and post-flood trauma lasts 6–12 months without support.

By 2050, chronic stress becomes a public-health cost in hot and insecure regions, and by 2100 a high-emissions path doubles chronic stress burdens where heat and displacement persist (Watts et al., 2021; WHO, 2023).

Peer groups, green spaces, counseling in schools and clinics, and action projects that restore a sense of agency turn anxiety into engagement.

Table 20.1 Future scenarios and predictions for climate anxiety, trauma and building psychological resilience (2030-2100)

Horizon	What changes for this chapter
2030	anxiety 35–45% of teens weekly
2050	chronic stress raises health costs
2100	chronic burden ×2 in hot, insecure zones

Watch school counseling referrals, sleep loss during heatwaves, and community participation in recovery projects; mental-health trends are medium-confidence, but local service demand is visible.

CHAPTER 21

GOVERNANCE, JUSTICE AND CIVIC ENGAGEMENT

Theme: climate governance, justice, equity and civic engagement

This chapter examines climate governance, justice, equity and civic engagement. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

21.1 International Frameworks and Paris Agreement

As Diana Liverman (2020) argues, climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For international frameworks and paris agreement within climate governance, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. In the terms of international frameworks and paris agreement, the claim shifts attention from prediction to preparation.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~100 million people are affected by climate disasters annually (IFRC, 2022). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. For courts, legislatures and civic organisations, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Averages describe the size of the problem, not who carries it. Decisions taken by courts, legislatures and civic organisations — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to

solve. Within courts, legislatures and civic organisations, the implication is a shorter list of decisions than the analysis suggests.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as courts, legislatures and civic organisations is concerned, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~100 million people are affected by climate disasters annually (IFRC, 2022). None of those signals is a perfect measure, but together they indicate direction. In the work of courts, legislatures and civic organisations, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. As far as courts, legislatures and civic organisations is concerned, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For courts, legislatures and civic organisations, the operational reading is to record which limit applies to each decision before the decision is taken.

21.2 National Adaptation Plans and Local Governance

As Mark Howden (2020) argues, adapting Agriculture to Climate Change)

Howden's statement comes from his 2007 PNAS paper 'Adapting agriculture to climate change' and his role as IPCC WGII Vice-Chair. 'Dealing with barriers to effective adaptation will require a comprehensive and dynamic policy approach' means technical options alone fail when institutions, markets, and knowledge systems are misaligned. 'Science, too, has to adapt' calls for linking scientific, local, and Indigenous knowledge, and for research that is co-produced with farmers and communities. Barriers he identifies include fragmented governance, short-term incentives, and poor access to tailored forecasts and finance. Howden emphasizes iterative learning — piloting, monitoring, and adjusting — rather than one-off plans. For national adaptation plans and local governance within climate governance, this directs attention to enabling conditions for adoption: trusted extension, tenure security, and policy that supports

diversification and ecosystem-based practices alongside technology. For national adaptation plans and local governance, the practical consequence is to name the decision that would have to change.

The aggregate picture hides the distribution, and in national adaptation plans and local governance the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within courts, legislatures and civic organisations, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. As far as courts, legislatures and civic organisations is concerned, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

Anyone deciding about national adaptation plans and local governance is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of courts, legislatures and civic organisations, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. For courts, legislatures and civic organisations, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Current reporting adds detail rather than a new direction: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). On its own it proves little; placed against the trend it shows the direction of travel. In the work of courts, legislatures and

civic organisations, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within courts, legislatures and civic organisations, the question is which of these decisions can be taken before the next event rather than during it.

21.3 Carbon Pricing and Climate Finance

Francesco La Camera (2023) puts it plainly: renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. Applied to carbon pricing and climate finance, the argument sets a test: does the measure still work when capacity is stretched?

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. As far as courts, legislatures and civic organisations is concerned, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The relevant question for carbon pricing and climate finance is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of courts, legislatures and civic organisations, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. For courts, legislatures and civic organisations, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: economic losses from climate disasters reach

~$200–300 billion per year (World Bank, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. Within courts, legislatures and civic organisations, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

A recent marker points the same way: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). Taken alone the episode is weather; taken with the record it is a data point on a slope. For courts, legislatures and civic organisations, the report is a reminder that the response is being designed against a target that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as courts, legislatures and civic organisations is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

21.4 Adaptation Finance Gaps and Investment

As Inger Andersen (2023) argues, adaptation finance gaps are widening, leaving the vulnerable to pay with lives.

UNEP's adaptation-gap reports compare finance needs with flows actually disbursed. The gap has widened every year the series has been published. Adaptation remains the smaller share of climate finance by a wide margin. For adaptation finance gaps and investment, the bearing is direct: the constraint is institutional as much as physical.

Any assessment of adaptation finance gaps and investment will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of courts, legislatures and civic organisations, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The economics of adaptation finance gaps and investment are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For courts, legislatures and civic organisations, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against

one budget year is often cheap against ten.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. Within courts, legislatures and civic organisations, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in adaptation finance gaps and investment depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as courts, legislatures and civic organisations is concerned, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). The item is useful because it fixes a date to a trend that otherwise drifts. Within courts, legislatures and civic organisations, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of courts, legislatures and civic organisations, the question is which of these decisions can be taken before the next event rather than during it.

21.5 Case Study: Governance, Justice and Civic Engagement

The case that anchors this chapter is documented and recent: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. Within courts, legislatures and civic organisations, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which courts, legislatures and civic organisations controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

21.6 Future Scenarios and Predictions: Governance, Justice and Civic Engagement

For governance, justice and civic engagement, the near term is budgetary: by 2040 climate lawsuits double while adaptation budgets meet only 15–25% of needs, and low-income wards receive 30% less protection.

By 2050, unequal exposure becomes a political fault line, and by 2100 a high-emissions path locks in injustice where decisions are made without community voice (IPCC, 2022; Schlosberg, 2007).

Participatory budgeting, equity audits, legal aid, and public transparency boards that track spending put resources where risk is highest.

Table 21.1 Future scenarios and predictions for climate governance, justice, equity and civic engagement (2030-2100)

Horizon	What changes for this chapter
2030	adaptation budgets 15–25% of needs
2050	unequal exposure becomes political
2100	injustice locked in without voice

Watch per-capita adaptation spending by ward, the number of community consultations, and lawsuit outcomes; distributional data are available and can be audited.

CHAPTER 22

CASE STUDIES: HEAT, FIRE, FLOOD AND URBAN EXTREMES

Theme: regional case studies of heat, wildfire, flood and urban climate extremes

The subject of this chapter is regional case studies of heat, wildfire, flood and urban climate extremes. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

22.1 Heat in South Asia and Urban Heat in Global Cities

As Friederike Otto (2021) argues, the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that heat in south asia and urban heat in global cities extremes are not random bad luck but part of a detectable trend. For regional case studies of heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. The implication for heat in south asia and urban heat in global cities is a question of sequence: what has to be in place before the event, not after it.

Uncertainty in heat in south asia and urban heat in global cities is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of urban authorities in fire- and flood-prone regions, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about heat in south asia and urban heat in global cities are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For urban authorities in fire- and flood-prone regions, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind heat in south asia and urban heat in global cities is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within urban authorities in fire- and flood-prone regions, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The news cycle supplies a concrete marker: Wildfires in Canada in 2023 burned ~15 million ha, more than double the previous record (1989, 6.7 Mha), sending smoke across North America (CIFFC, Nature 2024). Read against the longer record, it is an illustration rather than a proof. In the work of urban authorities in fire- and flood-prone regions, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. In the work of urban authorities in fire- and flood-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

22.2 Wildfires in Australia and North America

Friederike Otto (2021) puts it plainly: the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that

wildfires in australia and north america extremes are not random bad luck but part of a detectable trend. For regional case studies of heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. In the terms of wildfires in australia and north america, the claim shifts attention from prediction to preparation.

Discussion of wildfires in australia and north america usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Read that way, the question for urban authorities in fire- and flood-prone regions is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For urban authorities in fire- and flood-prone regions, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for wildfires in australia and north america is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within urban authorities in fire- and flood-prone regions, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

An honest account of wildfires in australia and north america distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of urban authorities in fire- and flood-prone regions, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Current reporting adds detail rather than a new direction: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). The value of such reports is that they date the change rather than explain it. For urban authorities in fire- and flood-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. For urban authorities in fire- and flood-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

22.3 Floods in Southeast Asia and Urban Flooding

As Chuang Zhao (2017) argues, each degree of warming lowers staple yields by ~6%, with wheat most vulnerable to heat.

Zhao and colleagues combined field warming experiments with models to estimate yield responses per degree. Staples fall by roughly six per cent per degree of local warming, with wheat most sensitive. The estimate applies to present-day locations and practices. For floods in southeast asia and urban flooding, the practical consequence is to name the decision that would have to change.

The record on floods in southeast asia and urban flooding is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how urban authorities in fire- and flood-prone regions should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within urban authorities in fire- and flood-prone regions, the test is whether the arrangement survives contact with a bad week.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

Not all statements about floods in southeast asia and urban flooding carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: sea level has risen ~20–25

cm since 1900 (IPCC, 2021). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of urban authorities in fire- and flood-prone regions, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and floods in southeast asia and urban flooding is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For urban authorities in fire- and flood-prone regions, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Bangladesh's Sidr-like Cyclone Sitrang (Oct 2022) inundated 10 million coastal residents (Bangladesh BMD, UN OCHA). It matters less as a record than as evidence that the baseline has already moved. Within urban authorities in fire- and flood-prone regions, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within urban authorities in fire- and flood-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

22.4 Storms and Hurricanes in the Caribbean

As Ana Vicedo-Cabrera (2021) argues, over one-third of heat deaths are attributable to anthropogenic warming — a burden we can already count.

Vicedo-Cabrera and colleagues attributed a large share of warm-season heat deaths to anthropogenic warming using multi-country mortality records. The finding converts a global average into a body count. Attribution of this kind depends on the choice of baseline period. Applied to storms and hurricanes in the caribbean, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The strength of the claim about storms and hurricanes in the caribbean varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). A central estimate without its range is not more precise, only less honest. In the work of urban authorities in fire- and flood-prone regions, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question storms and hurricanes in the caribbean is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For urban authorities in fire- and flood-prone regions, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on storms and hurricanes in the caribbean rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within urban authorities in fire- and flood-prone regions, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). Treated as a single observation it invites over-reading; treated as one of many it is informative. As far as urban authorities in fire- and flood-prone regions is concerned, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as urban authorities in fire- and flood-prone regions is concerned, the question is which of these decisions can be taken before the next event rather than during it.

22.5 Case Studies: Heat, Fire, Flood and Urban Extremes

One documented episode makes the argument concrete: Wildfires in Canada in 2023 burned ~15 million ha, more than double the previous record (1989, 6.7 Mha), sending smoke across North America (CIFFC, Nature 2024). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. As far as urban

authorities in fire- and flood-prone regions is concerned, the record supplies both the warning and the timetable.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence urban authorities in fire- and flood-prone regions can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

22.6 Future Scenarios and Predictions: Case Studies — Heat, Fire, Flood and Urban Extremes

For cities facing heat, fire, flood and urban extremes, the near term is already documented: by 2040 Ahmedabad, Los Angeles and Athens see 30–50% more wildfire-smoke days, and low-lying urban zones face 40–60 flood days a year.

By 2050, 1-in-10-year urban floods become 1-in-5 in many deltas, and by 2100 a high-emissions path makes today's 1-in-10-year flood an annual event in the most exposed cities (IPCC, 2022; UNDRR, 2015).

Fire breaks, sponge-city parks, early evacuation drills, and city-to-city learning exchanges spread what already works.

Table 22.1 Future scenarios and predictions for regional case studies of heat, wildfire, flood and urban climate extremes (2030-2100)

Horizon	What changes for this chapter
2030	smoke days +30–40%; flood days +20
2050	1-in-10 urban flood → 1-in-5
2100	1-in-10 flood becomes annual

Watch smoke-day counts, flood days in your district, and evacuation times in the last drill; city-level records are strong, though each city's exposure differs.

CHAPTER 23

CASE STUDIES: DROUGHT, FOOD, WATER AND SECURITY

Theme: regional case studies of drought, food crisis, water stress and security

This chapter deals with regional case studies of drought, food crisis, water stress and security. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

23.1 Drought in the Horn of Africa and Sahel

Tim Wheeler (2013) puts it plainly: food shocks travel through prices and nutrition, striking import-dependent poor first.

Wheeler and von Braun trace how food shocks travel through prices, trade and nutrition. Import-dependent poor households absorb the first and deepest losses. Short-term food aid does not prevent the next shock. Read against drought in the horn of africa and sahel, the point is that intention without capacity changes little.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~2 billion people lack safe water (WHO & UNICEF, 2023). As far as rural authorities in drought-prone regions is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). For that reason, average-based targets can be met while the most exposed end up worse off. In the work of rural authorities in drought-prone regions, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For rural authorities in drought-prone regions, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~2 billion people lack safe water (WHO & UNICEF, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within rural authorities in drought-prone regions, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The news cycle supplies a concrete marker: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). On its own it proves little; placed against the trend it shows the direction of travel. For rural authorities in drought-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. As far as rural authorities in drought-prone regions is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

23.2 Food Crisis in Sub-Saharan Africa and Price Volatility

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For food crisis in sub-saharan africa and price volatility within regional case studies of drought, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. The implication for food crisis in sub-saharan africa and price volatility is a question of sequence: what has to be in place before the event, not after it.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Because disadvantages compound, the

same event can be an inconvenience for one family and a permanent loss for another. In the work of rural authorities in drought-prone regions, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in food crisis in sub-saharan africa and price volatility depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~800 million people face food insecurity (FAO, 2023). For rural authorities in drought-prone regions, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~2 billion people lack safe water (WHO & UNICEF, 2023). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within rural authorities in drought-prone regions, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Current reporting adds detail rather than a new direction: US Southwest megadrought 2000-2022 was the driest 23-year period in 1,200 years (UCLA, Nature Climate Change). Taken alone the episode is weather; taken with the record it is a data point on a slope. Within rural authorities in drought-prone regions, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. In the work of rural authorities in drought-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

23.3 Water Stress in the Middle East and Groundwater Depletion

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For water stress in the middle east and groundwater depletion within regional case studies of drought, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. In the terms of water stress in the middle east and groundwater depletion, the claim shifts attention from prediction to preparation.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~800 million people face food insecurity (FAO, 2023). For rural authorities in drought-prone regions, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~2 billion people lack safe water (WHO & UNICEF, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within rural authorities in drought-prone regions, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~800 million people face food insecurity (FAO, 2023). Averages describe the size of the problem, not who carries it. Decisions taken by rural authorities in drought-prone regions — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. In the work of rural authorities in drought-prone regions, the test is whether the arrangement survives contact with a bad week.

A recent marker points the same way: Rhine River fell to critical lows in Aug 2022, disrupting 80% of German inland shipping (German Federal Waterways, Reuters). The item is useful because it fixes a date to a trend that otherwise drifts. As far as rural authorities in drought-prone regions is concerned, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For rural authorities in drought-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

23.4 Migration-Conflict Links in Fragile States

Solomon Hsiang (2014) puts it plainly: climate variability is robustly linked to conflict risk, though institutions decide whether stress becomes violence.

Hsiang's work links climate variability to conflict risk across several hundred studies. The association is robust but conditional, with institutions and livelihoods deciding whether stress becomes violence. Climate is a risk multiplier, not a cause. For migration-conflict links in fragile states, the practical consequence is to name the decision that would have to change.

The economics of migration-conflict links in fragile states are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~2 billion people lack safe water (WHO & UNICEF, 2023). Within rural authorities in drought-prone regions, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale

(UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The aggregate picture hides the distribution, and in migration-conflict links in fragile states the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~800 million people face food insecurity (FAO, 2023). In the work of rural authorities in drought-prone regions, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~2 billion people lack safe water (WHO & UNICEF, 2023). What the record does show is that authority, money and information have to move together; any one without the others stalls. For rural authorities in drought-prone regions, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: Amazon drought in late 2023 lowered Rio Negro to 120-year low, isolating 500,000 people (Brazil INPE, Reuters). What the report adds is timing, not a new mechanism. In the work of rural authorities in drought-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within rural authorities in drought-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

23.5 Case Studies: Drought, Food, Water and Security

A single episode illustrates the mechanism: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. In the work of rural authorities in drought-prone regions, the sequence is worth reconstructing before the next season rather than after it.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted

savings — a sequence rural authorities in drought-prone regions can measure in its own records.

Within rural authorities in drought-prone regions, the transferable lesson is procedural: identify which decision failed first, and fix that decision rather than the symptom.

23.6 Future Scenarios and Predictions: Case Studies — Drought, Food, Water and Security

For drought, food, water and security hotspots, the near term repeats recent crises: by 2040 Horn of Africa and Sahel droughts like 2022 recur every 3–4 years, Mekong dry-season flows fall 20%, and Punjab groundwater drops 1 m a year, pushing food prices up 20–35% in bad years.

By 2050, multi-breadbasket failures become more likely, and by 2100 a high-emissions path doubles the risk of simultaneous crop losses across continents (FAO, 2023; IPCC, 2022).

Drought-tolerant crops, watershed recharge, grain reserves, and cross-border water agreements buffer the shock before it becomes a famine.

Table 23.1 Future scenarios and predictions for regional case studies of drought, food crisis, water stress and security (2030-2100)

Horizon	What changes for this chapter
2030	drought every 4–5 years; prices +15%
2050	multi-breadbasket risk rises
2100	simultaneous crop loss risk ×2

Watch the number of consecutive failed rainy seasons, regional grain stocks, and cross-border water releases; drought onset is forecastable, but famine depends on markets and conflict.

CHAPTER 24

ACTION PLANS AND CHECKLISTS FOR SURVIVAL

Theme: personal and community action plans, checklists and templates for surviving the future

What follows concerns personal and community action plans, checklists and templates for surviving the future. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

24.1 30-Day Personal Action Plan: Water, Food, Shelter

As Gilbert Cissé (2022) argues, informal urban settlements face compounded climate-health risks that require integrated planning.

Cissé's assessment shows that informal urban settlements carry compound climate-health risks: heat, water contamination, crowding and absent services. Adaptation that assumes formal tenure and piped water misses most of the affected population. Planning has to start from the settlement, not the grid. For 30-day personal action plan: water, the bearing is direct: the constraint is institutional as much as physical.

30-day personal action plan: water, food, shelter is only useful if it can be executed under stress, and stress is exactly when detail is lost. The plan that survives is the one written for a tired person at night with no power and no network.

The common failure is not ambition but vagueness. 'Prepare a kit' means nothing at 2 a.m.; 'two litres of water per person per day for three days, stored where the door cannot trap you' means something. Specificity is the difference between a document and a capability.

Review matters more than design. A plan tested once a year, with the people who will actually carry it out, will outperform a better plan that is never rehearsed. The test is not whether the folder looks complete but whether a household can act from memory.

24.2 1-Year Community Action Plan: Governance and Networks

As Peter Gleick (2020) argues, water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. Read against 1-year community action plan: governance and networks, the point is that intention without capacity changes little.

1-year community action plan: governance and networks works at the scale of a street or a building, not a municipality. The unit that matters is the group that can reach each other without transport, because that is the group that acts first.

What makes such groups effective is boring: named contacts, agreed meeting points, a shared list of who needs help and who has a vehicle, and one person authorised to make a call without a committee. Trust is built by doing small things together before it is needed.

Sustainability depends on rotating responsibility. Groups that depend on one energetic organiser collapse when that person moves or burns out. Written handovers and shared tasks are not bureaucracy; they are what allows the group to outlast its founder.

24.3 Emergency Checklists: Go-Bags, Documents, Communication

Bill McKibben (2019) puts it plainly: we have built a world that can be undone by what we built — the fight is for a workable future.

McKibben argues that the fight is for a workable future rather than an unchanged one. The book catalogues movement successes and failures since 2007. Its purpose is strategic, not predictive. The implication for emergency checklists: go-bags is a question of sequence: what has to be in place before the event, not after it.

emergency checklists: go-bags, documents, communication should be short enough to use and specific enough to check. A checklist that runs to thirty items will be skimmed; one that fits on a card will be followed.

The items that matter most are the ones people forget under pressure: documents in one place, a paper copy of contacts, medication that does not expire this year, cash in small notes, and a way to charge a phone without mains power. Each is cheap; each is routinely missing.

Checklists also serve a social function. Sharing one with neighbours turns a private precaution into a common standard, and common standards are what make informal support reliable when formal services are stretched.

24.4 Household Templates: Risk Matrix and Water Audit

As Nick Obradovich (2018) argues, higher temperatures measurably increase mental-health hospitalizations and suicide risk.

Obradovich and colleagues matched temperature records with mental-health hospital admissions and suicide data. Both rise with heat, and the effect persists beyond a single day. The mechanism runs through sleep, medication and social contact. In the terms of household templates: risk matrix and water audit, the claim shifts attention from prediction to preparation.

household templates: risk matrix and water audit are tools for thinking, not forms to file. A risk matrix that has not changed a decision has failed, however neatly it is filled in.

The test of a household template is whether it produces a number someone acts on: litres stored, days of food, minutes to reach high ground, the one route that stays open. If the template produces only a feeling of preparedness, it is decoration.

Templates should be revised after every real event and every near miss. The revision is the learning; the document is only its memory.

24.5 Case Study: Action Plans and Checklists for Survival

The case that anchors this chapter is documented and recent: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. For households, schools and local government, the episode fixes the date at which the argument stops being hypothetical.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which households, schools and local government controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

24.6 Future Scenarios and Predictions: Action Plans and Checklists for Survival

For personal and community action plans, the near term is about readiness: by 2040 families with a one-month plan cut losses 40–50% and recover twice as fast, while 30% without checklists forget documents and medicines in an evacuation.

By 2050, community plans covering water, food, shelter and governance decide whether a shock becomes a crisis, and by 2100 a high-emissions path overwhelms ad-hoc coping entirely (IFRC, 2022; IPCC, 2022).

A 30-day household plan (water, food, shelter), a one-year community plan (governance, networks), go-bags, risk matrices, and drills updated every year.

Table 24.1 Future scenarios and predictions for personal and community action plans, checklists and templates for surviving the future (2030-2100)

Horizon	What changes for this chapter
2030	1-month plan cuts losses 40–50%
2050	community plans decide outcomes
2100	ad-hoc coping overwhelmed (high path)

Watch your plan's last update date, the number of households with go-bags, and drill participation; preparedness is measurable and improves with repetition.

APPENDIX A: LAMPIRAN (CHECKLISTS & PANDUAN)

Master Emergency Checklist

- Item 1: Verify supplies, documents, and communication plans.
- Item 2: Verify supplies, documents, and communication plans.
- Item 3: Verify supplies, documents, and communication plans.
- Item 4: Verify supplies, documents, and communication plans.
- Item 5: Verify supplies, documents, and communication plans.
- Item 6: Verify supplies, documents, and communication plans.
- Item 7: Verify supplies, documents, and communication plans.
- Item 8: Verify supplies, documents, and communication plans.
- Item 9: Verify supplies, documents, and communication plans.
- Item 10: Verify supplies, documents, and communication plans.

Panduan Penulisan dan Debat Akademik

Buku ini dirancang untuk digunakan dalam diskusi akademik. Saat memperdebatkan atau memperluas klaimnya:

- Selalu sitasi sumber primer (paper peer-reviewed, laporan IPCC).
- Bedakan antara observasi, atribusi, dan proyeksi.
- Akui ketidakpastian dan tingkat kepercayaan.
- Pertimbangkan berbagai jalur sosioekonomi (SSP).
- Sertakan dimensi keadilan dan kesetaraan.
- Perbarui klaim seiring munculnya bukti baru.

APPENDIX B: GLOSARIUM

- Adaptation: Adjusting to actual or expected climate impacts.
- Mitigation: Reducing greenhouse gas emissions.
- Resilience: Capacity to absorb disturbance and reorganize.
- Vulnerability: Susceptibility to harm.
- Exposure: Presence of people or assets in hazard-prone areas.
- Sensitivity: Degree to which a system is affected by climate.
- Adaptive capacity: Ability to adjust to climate change.
- Tipping point: Threshold beyond which change becomes self-sustaining.
- Greenhouse effect: Trapping of heat by atmospheric gases.
- Carbon footprint: Total greenhouse gas emissions caused by an entity.
- SSP: Shared Socioeconomic Pathway (scenario framework).
- RCP: Representative Concentration Pathway.
- Committed warming: Warming locked in by past emissions.

APPENDIX C: INDEKS

- Adaptation
- Biodiversity
- Carbon footprint
- Carbon pricing
- Climate finance
- Committed warming
- Conflict
- Drought
- Early warning
- Ecosystem
- Equity
- Exposure
- Flood
- Food security
- Greenhouse effect
- Health
- Heat wave
- Indigenous knowledge
- Infrastructure
- Justice
- Mental health
- Migration
- Mitigation
- Mutual aid
- RCP
- Renewable energy
- Resilience
- SSP
- Sea level rise
- Sensitivity
- Social capital
- Tipping point
- Vulnerability
- Water security

APPENDIX D: REFERENCES

- Adger, W.N. (2006). Vulnerability. Global Environmental Change, 16(3), 268-281.
- Aldrich, D.P. & Meyer, M.A. (2015). Social capital and community resilience. American Behavioral Scientist, 59(2), 254-269.
- Andersen, Inger. 2023. Adaptation Gap Report 2023. UNEP, Nairobi.
- Anderson, Kevin. 2020. Carbon budgets and fossil exploration critiques. Tyndall Centre, Manchester.
- Berkes, F. & Ross, H. (2013). Community resilience: toward an integrated approach. Society & Natural Resources, 26(1), 5-20.
- Birol, Fatih. 2023. World Energy Outlook 2023. International Energy Agency, Paris.
- Burke, M., Hsiang, S.M. & Miguel, E. (2015). Global non-linear effect of temperature on economic production. Nature, 527, 235-239.
- Canadell, Pep et al. 2021. Global Carbon Project: Fossil Emissions Trends. Global Carbon Project.
- Cissé, Gilbert et al. 2022. Health, Wellbeing and the Changing Structure of Communities. IPCC AR6 WG2 Chapter 7. Cambridge University Press.
- Clayton, S. (2020). Climate anxiety: Psychological responses to climate change. Journal of Anxiety Disorders, 74, 102263.
- Cook, John et al. 2013. Quantifying the consensus on anthropogenic global warming. Environmental Research Letters, 8(2), 024024.
- Cutter, S.L. et al. (2008). A place-based model for understanding community resilience. Global Environmental Change, 18(4), 598-606.
- Dessler, Andrew. 2021. Introduction to Modern Climate Change. Cambridge University Press.
- Ebi, K.L. et al. (2021). Hot weather and heat extremes: health risks. The Lancet, 398(10301), 698-708.
- FAO. (2021). Climate Change: Unpacking the Burden on Food Safety. FAO, Rome.
- FAO. (2022). The State of the World's Land and Water Resources (SOLAW 2021). FAO, Rome.
- FAO. (2023). The State of Food Security and Nutrition in the World. FAO, Rome.
- Folke, C. et al. (2010). Resilience thinking. Ecology and Society, 15(4), 20.
- Friedlingstein, Pierre et al. 2023. Global Carbon Budget 2023: carbon cycle and sinks. Earth System Science Data.
- Ghosh, Amitav. 2016. The Great Derangement: Climate Change and the Unthinkable. University of Chicago Press.
- Gleick, Peter H. 2023. Water, Climate and Governance Crises. Pacific Institute, Oakland.
- Gore, Al. 2006. An Inconvenient Truth: The Planetary Emergency of Global Warming and What We Can Do About It. Rodale Press.
- Haigh, Joanna D. 2020. Solar Variability and Climate. Imperial College Press.

- Hansen, J. et al. (2016). Ice melt, sea level rise and superstorms: evidence from paleoclimate data, climate modeling, and modern observations that 2°C global warming could be dangerous. Atmos. Chem. Phys., 16, 3761-3812.
- Hausfather, Zeke et al. 2023. Berkeley Earth Global Temperature Report 2023. Berkeley Earth.
- Hayhoe, K. (2021). Saving Us: A Climate Scientist's Case for Hope and Healing in a Divided World. One Signal Publishers.
- Hickman, C. et al. (2021). Climate anxiety in children and young people. The Lancet Planetary Health, 5(12), e863-e873.
- Howden, S.M. et al. (2007). Adapting agriculture to climate change. PNAS, 104(50), 19691-19696.
- Hsiang, S.M. & Burke, M. (2014). Climate, conflict, and social stability. Climatic Change, 123, 39-55.
- Huq, S. (2001-2023). ICCCAD / IIED oral statements; COP interventions archived via IPCC AR3-AR5 lead authorship and Nature obituary 2023.
- IEA. (2023). World Energy Outlook 2023. International Energy Agency, Paris.
- IFRC. (2022). World Disasters Report 2022. IFRC, Geneva.
- IMF. (2022). World Economic Outlook: War Sets Back Global Recovery. IMF, Washington DC.
- IPBES. (2019). Global Assessment Report on Biodiversity and Ecosystem Services. IPBES Secretariat, Bonn.
- IPCC. (2018). Global Warming of 1.5 C (SR15). IPCC, Geneva.
- IPCC. (2019). Climate Change and Land (SRCCL). IPCC, Geneva.
- IPCC. (2019). Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC). IPCC, Geneva.
- IPCC. (2021). Climate Change 2021: The Physical Science Basis (AR6 WG1). Cambridge University Press.
- IPCC. (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability (AR6 WG2). Cambridge University Press.
- IPCC. (2022). Climate Change 2022: Mitigation of Climate Change (AR6 WG3). Cambridge University Press.
- IPCC. (2023). Climate Change 2023: Synthesis Report. IPCC, Geneva.
- IRENA. (2023). World Energy Transitions Outlook 2023. IRENA, Abu Dhabi.
- Jacobs, Katharine et al. 2014. Climate Change Impacts in the United States: The Third National Climate Assessment. U.S. Global Change Research Program.
- Klein, Naomi. 2014. This Changes Everything: Capitalism vs. The Climate. Simon & Schuster.
- Klinsky, Sonja & Brankovic, Jasmina. 2018. The Legacies of Climate Change. Routledge.
- Kolbert, Elizabeth. 2014. The Sixth Extinction: An Unnatural History. Henry Holt and Co.
-

Kulp, S.A. & Strauss, B.H. (2019). New elevation data triple estimates of global vulnerability to sea-level rise. Nature Communications, 10, 4844.
- La Camera, Francesco. 2023. World Energy Transitions Outlook 2023. IRENA, Abu Dhabi.
- Le Quéré, Corinne et al. 2023. Global Carbon Budget 2023. Earth System Science Data, 15(12), 5301-5369.
- Liverman, D.M. (2013-2023). Climate justice scholarship; Arizona House Climate Crisis testimony 2019 & IPCC author work.
- Lobell, D.B., Schlenker, W. & Costa-Roberts, J. (2011). Climate trends and global crop production since 1980. Science, 333(6042), 616-620.
- Mach, K.J. et al. (2019). Climate as a risk factor for armed conflict. Nature, 571, 193-197.
- Mann, M.E., Bradley, R.S. & Hughes, M.K. (1999). Northern Hemisphere temperatures during the past millennium. Geophys. Res. Lett., 26(6), 759-762 (hockey stick).
- McKibben, Bill. 2019. Falter: Has the Human Game Begun to Play Itself Out? Henry Holt and Co.
- Mizutori, Mami. 2019. Sendai Framework for Disaster Risk Reduction Implementation. UNDRR, Geneva.
- NASA. (2024). Global Climate Change: Vital Signs. climate.nasa.gov.
- Neira, M. (2021). Foreword/press statements COP26 Special Report on Climate Change and Health. WHO, Geneva.
- NOAA. (2024). State of the Climate. ncei.noaa.gov.
- Obradovich, N. et al. (2018). Empirical evidence of mental health risks posed by climate change. PNAS, 115(43), 10953-10958.
- OECD. (2021). Managing Climate Risks, Facing Essential Challenges. OECD, Paris.
- OECD. (2023). Climate Finance Provided and Mobilised by Developed Countries. OECD, Paris.
- Oppenheimer, M. et al. (2019). Sea Level Rise and Implications for Low-Lying Islands, Coasts and Communities. SROCC Chapter 4.
- Oreskes, Naomi & Conway, Erik M. 2010. Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Global Warming. Bloomsbury Press.
- Otto, F.E.L. et al. (2021). Attribution of the 2021 Pacific Northwest heatwave — virtually impossible without human-caused climate change. World Weather Attribution.
- Patel, Raj. 2022. Food Systems, Power and Hunger in a Warming World. Verso.
- Peters, Glen P. et al. 2022. Emissions Gap and Carbon Budget Perspectives. CICERO, Oslo.
- Pörtner, H.-O. et al. (2022). IPCC AR6 WGII Summary for Policymakers press conference statements 28 Feb 2022.

- Red Cross. (2020). Heat Waves: A Silent Killer. IFRC, Geneva.
- Roberts, D. et al. (2022). IPCC AR6 WGII Chapter 6 & press statements — adaptation and mitigation pursued with equal force.
- Robinson, Kim Stanley. 2020. The Ministry for the Future. Orbit Books.
- Rockström, J. et al. (2009). A safe operating space for humanity. Nature, 461, 472-475; Richardson et al. (2023). Planetary boundaries update. Sci. Adv., 9.
- Romanello, M. et al. (2023). The 2023 report of the Lancet Countdown. The Lancet, 402(10419), 2346-2394.
- Rosenzweig, C. et al. (2019). Special Report on Climate Change and Land; WFP Laureate remarks 2022 (multi-breadbasket failure risk).
- Roy, Joyashree et al. 2019. Sustainable Land Management and Climate. IPCC SRCCL Chapter 4.
- Schlosberg, D. (2007). Defining Environmental Justice: Theories, Movements, and Nature. Oxford University Press.
- Steffen, W. et al. (2018). Trajectories of the Earth System in the Anthropocene. PNAS, 115(33), 8252-8259.
- Stern, N. (2007). The Economics of Climate Change: The Stern Review. Cambridge University Press.
- Tschakert, Petra et al. 2021. Lived Experience of Climate Change: Heat, Flood and Limits. Wiley.
- UNDP. (2022). Human Development Report 2021/22. UNDP, New York.
- UNDRR. (2015). Sendai Framework for Disaster Risk Reduction 2015-2030. UNDRR, Geneva.
- UNEP. (2022). Emissions Gap Report 2022. UNEP, Nairobi.
- UNEP. (2023). Adaptation Gap Report 2023. UNEP, Nairobi.
- UNESCO. (2020). Education for Sustainable Development: A Roadmap. UNESCO, Paris.
- UNESCO. (2021). Getting Every School Climate-Ready. UNESCO, Paris.
- UNHCR. (2023). Climate Change and Displacement. unhcr.org.
- United Nations. (2015). 2030 Agenda for Sustainable Development. UN, New York.
- United Nations. (2015). Paris Agreement. unfccc.int.
- Vicedo-Cabrera, A.M. et al. (2021). Burden of heat-related mortality attributable to recent human-induced climate change. Nature Climate Change, 11, 492-500.
- Wallace-Wells, David. 2019. The Uninhabitable Earth: Life After Warming. Crown Publishing.
- Watts, N. et al. (2021). The 2021 report of the Lancet Countdown on health and climate change. The Lancet, 398(10311), 1619-1662.
- Wheeler, T. & von Braun, J. (2013). Climate change impacts on global food security. Science, 341(6145), 508-513.
- WHO & UNICEF. (2023). Progress on Household Drinking Water, Sanitation and Hygiene 2000-2022. JMP.

- WHO. (2021). COP26 Special Report on Climate Change and Health. WHO, Geneva.
- WHO. (2023). Climate Change and Health Fact Sheet. who.int.
- WMO. (2022). Early Warnings for All. WMO, Geneva.
- WMO. (2023). State of the Global Climate 2023. World Meteorological Organization, Geneva.
- World Bank. (2021). Groundswell: Acting on Internal Climate Migration. World Bank, Washington DC.
- World Bank. (2022). What You Need to Know About Food Security and Climate Change. World Bank, Washington DC.
- Zhao, C. et al. (2017). Temperature increase reduces global yields of major crops. PNAS, 114(35), 9326-9331.

APPENDIX E: ABOUT THE AUTHOR

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APPENDIX F: EXTENDED ANALYTICAL NOTES

Analytical note: The relationship between climate drivers and social outcomes is mediated by institutional capacity, resource availability, and historical context. No single factor determines the result.

Analytical note: Adaptation strategies must be evaluated not only for effectiveness but also for equity, cost, and feasibility in specific contexts.

Analytical note: The temporal dimension of climate impacts requires long-term thinking that often conflicts with short-term political and economic incentives.

Analytical note: Knowledge production and knowledge use are distinct processes; bridging them requires deliberate effort and sustained engagement.

Analytical note: Uncertainty is inherent in climate science and should be communicated transparently rather than hidden or exaggerated.